claude-windows / dbc4cdae-04f6-4d06-af90-397bbce3fc57

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\dbc4cdae-04f6-4d06-af90-397bbce3fc57\subagents\agent-ac396efe913fa3264.jsonl`
- SHA-256: `224f3ed95dc435acd974513b1b2c9f43b4b98ed5615cd59ddad60c8a8ba893a2`
- Source modified: `2026-07-06T20:15:48+00:00`
- Imported at: `2026-07-08T15:59:37+00:00`
- project: `subagents`
- session_id: `dbc4cdae-04f6-4d06-af90-397bbce3fc57`

Transcript

1. user (2026-07-06T20:13:55.756Z)

```
Read-only investigation. Do NOT modify any files. I need exact content from several docs in the
repo at C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer. For each item, give
me the EXACT file path and quote the concrete values (paths, URIs, filenames, resolutions, fps,
F1 numbers, commands) with the line region where they appear.

Read these files fully and extract the requested facts:

1. C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-
indexer\docs\GOLDEN_REGRESSION_FIXTURES.md
   - Where do golden eval VIDEOS live (local paths, GCS gs:// URIs)? List actual
filenames/paths.
   - Where are ground-truth LABELS and what format (CSV columns)? How are video<->label paired?
   - Any small subset (2-3 clips) named for a quick run, with resolution/fps.

2. C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\EVALUATION.md
   - The eval command(s) for a single video, and what inputs they need.
   - The F1/precision/recall definition used (rally-level? tolerance? IoU threshold?).

3. C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\docs\DATA_IN_GCS.md
   - GCS bucket names and gs:// URIs for golden videos and labels. List actual bucket/prefix
paths.

4. C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-
indexer\docs\data_pipeline\GOLDEN_VIDEOS.md and GOLDEN_SET_PHASE1_VIDEOS.md and
GOLDEN_SET_IMPLEMENTATION.md (all three, in docs\data_pipeline\)
   - Actual golden video filenames, GCS paths, resolutions, fps, and the label CSV format
(columns).
   - Video<->label pairing convention (e.g. X.MP4 <-> X.rallies.csv).
   - Names of 2-3 specific clips usable for a quick numbers run.

5. C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-
indexer\docs\R2_EVAL_METRIC_DESIGN.md
   - The exact definition of tolF1 vs F1, and the tau (?) tolerance values (in seconds or
pixels?).

Return a concise, structured report organized by the 4 questions: (a) golden video
locations/filenames/GCS URIs, (b) ground-truth label locations + CSV format + pairing, (c) 2-3
quick-run clips with resolution/fps, (d) F1/tolF1 definition + ?. Include exact file:line-region
evidence. Flag anything not found as an open question. Return findings as text only ? do not
write any files.
```

2. assistant (2026-07-06T20:14:01.491Z)

```
I'll investigate these documentation files. Let me start by reading all the requested files in
parallel.
```

3. user (2026-07-06T20:14:02.821Z)

1 # Golden Regression Fixtures ? video-free, non-attributable regression suite
2
3 > **Status:** `IN PROGRESS` ? owner-gated · **Owner:** Avi · **Created:** 2026-06-16 · **Last updated:** 2026-06-16 · **Legal:** owner **accepted the researched risk** 2026-06-16 (conditions §3 stand; this is the owner's informed acceptance, **not** a retained-counsel opinion)
4 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE, per archive policy)
5 > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`; pointer in memory `current-state` (and `docs/NEXT_STEPS.md` once work starts).
6 > **Relation to existing docs:** EXTENDS (does not supersede) `docs/GOLDEN_DATA_SHARING.md` (#175); reuses the eval stack documented in `docs/EVALUATION.md` / `docs/EXPERIMENT_HARNESS.md`; quality floor ties to `docs/QUALITY_ITERATIONS.md`.
7 > **Honesty note:** §3 (Legal) is **researched general information, NOT legal advice** ? it gates on retained counsel (§8). §2/§4/§7 mark which claims are **[verified]** against code vs **[design]** proposals.
8
9 ---
10
11 ## 0. TL;DR
12
13 Let a contributor on **another machine with no video, GPU, WSL, or DB** run the rally-segmentation regression suite from **committed structured files**, so refactors and non-quality PRs get a red/green side-effect signal. Two layers, built once and grown by accretion:
14
15 1. **Mechanics** ? freeze the deterministic pipeline at the **post-detector trajectory seam** and replay it through the *already-shipped* eval stack (`experiment.py` / `metrics.py` / `windowing.py` / `regression.py`), with two auto-discovered gates: a **characterization/snapshot** gate (catches any behaviour change) and a **quality-floor** gate (catches quality drops).
16 2. **Privacy/legal filter (SEALED-FIXTURES)** ? only **de-attributed, owner-source, biometric-free** numbers are ever committed; richer/sensitive streams stay key-gated or out-of-repo. Non-attributability comes from **deletion-at-source**, not encryption. ? "Minimized" means **stream minimization** (shuttle-only, no person/audio/pose) ? **NOT** reduction of the shuttle stream to ~5 scalars: the de-attributed per-frame `trajectory.csv` IS committed (see §4c + its risk-acceptance re-confirmation note).
17
18 **The convergence that makes this clean:** the *privacy* answer (publish shuttle-only, drop person/pose streams) and the *determinism* answer (freeze only the `CONTROL` path ? which runs **no numpy execution**, keeping the numpy/BLAS learned scorer off the cross-machine gate) point at the **same minimal public tier**. That coincidence is the spine of the design. (numpy is still *imported* transitively via `experiment` and must be installed; the CONTROL replay just never calls into it.)
19
20 **Two findings already actionable, independent of the rest:**
21 - ? **Live de-attribution bug** *(verified)*: real player/venue names are committed today in `eval_baselines/heuristic_lovo_n6.json`, `eval_baselines/gemini_wrapper_2026-06-10.json`, and `backend/eval/eval_partial_labels.py` (the last also has a personal `C:/Users/avidu/...` path) ? and in git history. ? **Deliverable P0.**
22 - ? **Determinism trap** *(verified, corroborated by two independent red-teams)*: the learned variants (`shuttle_only`, `fusion`) train a numpy/OpenBLAS `DYNAMIC_ARCH` logistic regression whose decisions flip near the 0.5 threshold across CPUs ? would false-RED on a contributor's machine. ? the cross-machine gate must be **CONTROL/static-only**.
23
24 ---
25
26 ## 1. Problem & goal
27
28 The rally-quality measurement is split across two machines by capability (`docs/GOLDEN_DATA_SHARING.md`):
29 - `backend/eval/fusion_golden.py` **extracts** `*_golden_features.json` ? needs the **video + trajectory CSV + labels + GPU/WSL** (GPU box only).
30 - `backend/eval/fusion_compare.py` / `backend/eval/ablation.py` **re-score** those JSONs

on **pure CPU, no video** ? any machine.

The CPU re-scoring is *already* video-free, but the inputs live in gitignored `output/` / private OneDrive, so the regression tests that depend on them **skip everywhere** (`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`). The committed `eval_baselines/` baselines exist but `regression.py` still scores DB-stored predictions (needs a pipeline run on the video).

Goal: commit a tiny, version-tagged, **numeric-only, non-attributable** per-fixture corpus + auto-discovered gates so a clean `git clone` / CI gets a real red/green regression signal with **zero video/GPU/DB**, while never leaking attributable or sensitive data.

---

2. Decisions locked

#	Decision	Source	Implication
D1	**Corpus is owner-recorded / owned**	owner answer 2026-06-16	Legal analysis takes **Fork A** (clean): no third-party copyright/ToS infringement; concern is trade-secret/moat + privacy. Strongest non-attributability (no public reference to correlate against).
D2	**Repo private now, may open-source later**	owner answer 2026-06-16	Design for **public-grade hygiene from day one** ? git history is permanent; nothing scrubable-later.
D3	**Adults, consent not formalized**	owner answer 2026-06-16	**Person/pose/gait streams stay OUT of the public tier** (DPDP/GDPR derived-personal-data risk without consent). Public tier = **shuttle + abstract intervals + non-biometric features only**.
D4	**Freeze seam = post-detector trajectory** *(design; SHIPPED differs ? see ?)*	workflow-1 synthesis	Widest deterministic surface; reuse the shipped eval stack. ? **Correction (audit 2026-06-16):** the shipped freeze **commits the de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** (shuttle-only, time-origin reset) ? it is **NOT** minimized to ~5 aggregate scalars. See §4c (and its risk-acceptance re-confirmation note).
D5	**Cross-machine gate is CONTROL/static-only** *(design)*	both red-teams [verified cause]	Learned variants (numpy/BLAS) are **owner-box/Tier-1 only**, asserted with tolerance, never as a cross-machine byte-snapshot.
D6	**Non-attributability = deletion-at-source, encryption = public-only barrier** *(design)*	workflow-2 synthesis	A contributor who runs the tests necessarily reads plaintext+key (DRM/analog-hole). So we **delete** attribution, not hide it. Encryption only stops the no-key public.
D7	**Two tiers** *(design)*	workflow-2 synthesis	**Tier-0** public/no-key (shuttle-only, de-attributed). **Tier-1** key-gated/out-of-repo (person/audio, full corpus), **skips cleanly when absent**.

---

3. Legal foundation (researched ? gates on counsel, §8)

Core verdict (settled across IN/US/EU): the derived numbers ? per-frame shuttle (x,y), rally start/end intervals, audio/court/aggregate-person feature values ? **do NOT inherit the source video's copyright.** They are uncopyrightable *facts/measurements*, not authored expression, and not a §101 "derivative work."

Jurisdiction	Copyright of the data	Database right	Contract / ToS	Net
India (primary)	Not copyrightable as facts (*EBC v. D.B. Modak*); but compilations get thin protection on a low "skill/labour" bar (*Burlington*, *OLX v. Padawan*) ? only bites if **lifted from a 3rd-party dataset**, not self-derived	**None** (no sui-generis right) ? a real permissiveness edge	Anti-scraping ToS (untested in court) + **IT Act s.43/s.66** civil/criminal exposure for 3rd-party ingestion	**Owned-source + minors-excluded + no person/pose = clean**
US	Not copyrightable (*Feist*; §102(b); *NBA v. Motorola*)	None (*Feist* killed sweat-of-the-brow)	**Dominant risk.** ToS enforceable where copyright/CFAA fail (*ProCD*, *hiQ* ? injunction + **forced deletion**). Remedy for redistribution is takedown, not nominal damages	Publishable as facts **iff** owned/permissioned source + no barring ToS + no person/biometric/minor stream

| **EU / UK** | Not copyrightable (*Football Dataco v. Yahoo*) | **Genuinely contestable.** *Football Dataco v. Sportradar*: recording pre-existing on-court facts = "**obtaining**", not "creating" ? our "we created it" premise is **overconfident**; harm test = *CV-Online Latvia* | Ryanair v. PR Aviation: ToS can ban extraction/redistribution even with no IP right | Get EU+UK advice before redistributing numeric streams at scale |

Provenance is the decisive fork. **Fork A (owner-recorded, unpublished)** ? the only path safe to commit publicly; collapses to a trade-secret *choice* (Indian breach-of-confidence / Contract Act 1872 / IT Act s.72 ? **not** US DTSA) + privacy. **Fork B (broadcast/YouTube/scraped)** ? do **not** commit; the extraction act needs a license or now-**contested** US fair use (*Thomson Reuters v. Ross* 2025; USCO May-2025), and unlawful acquisition independently sinks it (*Bartz v. Anthropic*, ~$1.5B settlement). Caveat: owner footage **shot at a third party's event** (tournament/club) carries ticket/venue/organiser/broadcaster terms ? treat as Fork B until cleared.

Publish-by-stream (privacy):
- ? **Publish-OK:** shuttle (x,y)+Visibility (inanimate object), abstract rally/GT **intervals** (identity-stripped), non-biometric audio/court/aggregate-person features detached from identity.
- ?? **Conditional (effectively never in v1):** person bounding boxes that *persistently single out a player* ? flips to GDPR Art.9 the moment it identifies. This architecture is prone to that flip ? treat per-player tracking as a tripwire.
- ? **Never publish:** pose/skeleton/**gait** (re-identifies at 80?95% even with faces blurred ? *Weiss et al. 2025*), face crops, any named-player?movement mapping. Matches the existing "gait/biometrics strictly future (GDPR Art.9)" guardrail.
- ? **Minors flip everything:** DPDP "child" = under 18; s.9(3) **prohibits behavioural monitoring** (the Rule-12 exemptions don't cover this project). A persistent per-player signature on a minor is, conservatively, prohibited. **Exclude minors** from any persisted stream beyond inanimate shuttle/interval data.

Non-attributability is relative, not absolute. A per-video trajectory is a fingerprint (de Montjoye: 4 points ? 95% re-ID). It's only non-attributable because the **Fork-A source stays unpublished** (no public reference to correlate against). Per *EDPS v. SRB* (CJEU C-413/23 P, 2025) it **remains personal data to the holder** who can re-run the detector. So: conditional, revocable (publish the source and the link snaps back), forward-looking (re-assess as the public-video environment grows). Never market it as "anonymous."

Key authorities: Feist 499 U.S. 340; NBA v. Motorola 105 F.3d 841; 17 U.S.C. §101/§102(b); Sega v. Accolade 977 F.2d 1510; Football Dataco v. Yahoo (C-604/10) & v. Sportradar (C-173/11); BHB v. William Hill (C-203/02); CV-Online Latvia (C-762/19); Ryanair v. PR Aviation (C-30/14); Thomson Reuters v. Ross (D. Del. 2025); EBC v. D.B. Modak; DPDP Act 2023 s.3/s.9 + Rules 2025; EDPS v. SRB (C-413/23 P). Full URLs in the research artifact (§10).

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4. Architecture (design)

4a. Freeze seam + reuse map
Freeze at the **trajectory CSV ? CONTROL-path replay**. One shared helper `backend/eval/golden_fixtures.py::replay_fixture()` is the *single* code path used by both the freeze tool and both gates. As **shipped**, that path is (per the module docstring):
`parse_trajectory_csv ? distill_local.build_candidates(proposal=(vt,mg,mwd)) ? VideoCandidates(5 shuttle cols only) ? experiment.predict(CONTROL) [keep all ? merge_overlaps] ? metrics.score`.
Reuses **verbatim**: `metrics.score/interval_iou`, `experiment.predict/CONTROL/VideoCandidates`, `distill_local.build_candidates`, `tracknet_runner.parse_trajectory_csv`/`trajectory_to_action_windows`, `rally_gate` net-crossings (inside `build_candidates`), `gt_loader.load_gt_intervals` (strict), `regression.gt_hash`, `eval_baselines/` committed-baseline precedent.
> ? **Correction (audit 2026-06-16):** the replay does **NOT** import `windowing.py` ? there is no `windowing.build_windows(preset)` / `WindowingPreset` / "default SERVED" call. The windowing params travel as a **raw 3-tuple `(vt, mg, mwd)`** pinned in each fixture's `provenance.json` and passed straight to `build_candidates`; the harness deliberately avoids the named `windowing.SERVED` constant so a future SERVED retune can't silently churn fixtures. (`segmentation_metrics`, `experiment.compare/Variant.spec_hash`, `regression.save_baseline`,

`eval_stats` are **not** on the M1 replay path either ? the quality-floor gate M4 will add the baseline/stats pieces.)
83
84 ### 4b. Tiered model (SEALED-FIXTURES)
85 | Tier | Where | Key? | Contents | Test behaviour |
86 |---|---|---|---|---|
87 | **Tier-0** | `eval_baselines/fixtures/` (tracked) | none | opaque surrogate slug, fps/frame_width (retained, NOT quantized), the **de-attributed per-frame `trajectory.csv`** `(Frame,Visibility,X,Y)` (shuttle-only, time-origin reset), the 5 shuttle feature columns inside `expected.json`, strict-slug `label`, relative `gts`. **No person/audio/pose.** ? NOT minimized to 5 scalars ? see §4c. | **Unconditional** ? runs on every PR/fork; turns today's skipped litmus into a real public gate |
88 | **Tier-1** | OneDrive (default) **or** SOPS-encrypted in-repo | token/age key | full corpus **with** person/audio; authoritative ?F1/CI/sign-test + exact Gate-A litmus | **Skips cleanly** when key/data absent (generalize `_golden_present()` ? `_tier1_present()`) |
89
90 A fresh no-key clone is always **green**; coverage never depends on a secret.
91
92 ### 4c. Anti-attribution measures (Tier-0)
93
94 > ? **RISK-ACCEPTANCE ? OWNER RE-CONFIRMED option (a), 2026-06-22.** The shipped freeze (`freeze_real_fixture`) commits the **de-attributed per-frame `(Frame,Visibility,X,Y)` `trajectory.csv`** (time-origin reset + opaque slug + metadata strip + scenario-only label) ? **NOT** minimized to ~5 scalars. A per-frame stream is a de Montjoye fingerprint, so non-attributability rests on the **source staying unpublished** (EDPS v. SRB), not on minimization. (An earlier draft wrongly claimed "5 scalars / never the per-frame CSV"; the audit corrected it 2026-06-16.) The owner cleared ALL owned golden videos (owned + minors-free) and accepted the corrected per-frame basis on 2026-06-22 ? the **6 real fixtures shipped under P3**. Owner alternative ? true ~5-scalar minimization ? remains available if these ever go public.
95
96 - **Opaque RANDOM surrogate id** (`os.urandom` + a private surrogate?source map kept off-git) ? *not* the MD5 `video_id`, and *not* HMAC(salt, video_id?name) (red-team: brute-forceable over a tiny known roster if the salt leaks). Never persist the source `name`/filename.
97 - **Strict-slug `label` + dropped `source_note`** ? the operator `label` is validated against `^[a-z0-9][a-z0-9_\-]*$` (a free-text identifier like a player/event name is refused), and operator free-text `source_note` is **accepted-but-NOT-persisted** (never reaches the committed `provenance.json`) ? both close the free-text identifier leak `_scan_forbidden` could not police (audit fix #1).
98 - **Metadata strip** ? no filename, video_id, match/event/player identity, capture date.
99 - **Time-origin reset** ? subtract per-fixture `t0` (6dp-rounded) from all start/end/gts. **Metric-invariant within `_FLOAT_TOL`** (`metrics.interval_iou` and start/end-error are pure differences): the genuine (raw) reset-vs-no-reset score delta sits at the float-error floor (~3.3e-7); bit-identical for whole-second offsets. See §4c-note below and §6.
100 - **Per-frame trajectory IS committed (de-attributed), NOT minimized to scalars** ? Tier-0 commits the de-attributed per-frame `trajectory.csv` (shuttle stream only). It is **not** reduced to ~5 aggregate scalars; the 5 shuttle *feature* columns appear (computed) inside `expected.json`, but the raw per-frame `(Frame,Visibility,X,Y)` stream is part of the committed fixture. (This is the corrected basis the ? note above gates on.)
101 - **Stream minimization (streams, not resolution)** ? Tier-0 keeps the shuttle block only; person/audio excluded; pose/gait/face have **no field in the schema by design** (positive `_scan_forbidden` assertion on the written bytes).
102 - **fps/frame_width are RETAINED as scale constants** ? they are NOT quantized today (quantization is a deferred P3 option, see §7 ?). Reconcile any "quantize fps/frame_width" wording elsewhere to this.
103 - **Provenance hard-fail gate** ? the de-attribution pass refuses to emit a public fixture for any record not tagged owner-recorded(Fork-A) + minors-excluded + non-blank-license, and refuses a negative reset label (timeline predates the trajectory). Writes go to a temp dir and are promoted only after the privacy scan passes, so a refused freeze leaves no partial fixture (audit fixes #3a/#3b).
104
105 ### 4d. The two gates
106 - **Characterization/snapshot** ? replay fixture, **parse-compare** JSON (never byte-compare ? CRLF/float-format safe; `core.autocrlf=true` here), exact on int/structural fields, `approx(abs=1e-6)` on floats; stamp + check windowing-preset/label/gt fingerprints (incommensurable ? loud fail). Catches any behaviour change even at constant F1. **CONTROL path

only** across machines.

- **Quality-floor** ? score vs committed labels; assert per-video + mean F1 ? floor in `eval_baselines/regression_baseline.json` (created here; carries a **windowing fingerprint** so a served-default change is flagged incommensurable, not silently compared); emit `eval_stats` paired-delta-CI + sign test (NUMBERS INFORM); add a **metrics-vs-base-branch** paired delta. Synthetic-tier floors measure **correctness, not field quality** (documented).

4e. Serialization & optional protobuf

JSON with `sort_keys=True` + pre-quantized floats + LF is sufficient. **Protobuf is optional (O1)** ? adopt only if cross-OS byte-stable snapshots are wanted (proto3 `rally.fixture.v1`, codegen + `protoc && git diff --exit-code` guard, `*.pb binary -text`). Protobuf is **encoding, not secrecy** (trivially `--decode_raw`).

4f. Encryption & keys (Tier-1 only)

Default = out-of-repo + token (existing OneDrive `RALLY_GOLDEN_DIR` seam ? zero sensitive bytes in git history, sidesteps the legal "redistribution" act). **Alternative = SOPS + age** (per-recipient, revocable, value-level diffs) for Fork-A-clean-but-private data wanting in-repo versioning. Rejected: git-crypt (whole-repo re-key), git-lfs (storage ? secrecy). CI secret exposed **only to the Tier-1 job on trusted branches, never to fork-PR runners**. The **HMAC/surrogate salt** (if used) and the surrogate?source map live with the private corpus, rotated per release. Document explicitly that Tier-1 crypto defends **only** the no-key public.

---

5. Threat model (who's protected vs not)

Actor	Protected against	NOT protected (by design / residual)
TM1 no-key public	map fixture?source/match/player; decrypt Tier-1; confirm a source even holding the candidate file	read Tier-0 numbers + run shuttle-only harness (intended; useless-in-isolation)
TM2 keyed contributor	recover filename/video_id/match/face/pose/gait ? *never written* (survives the analog hole)	read every value the suite reads (unpreventable; not pretended otherwise)
TM3 CI runner	leaked log can't leak attribution it never had	runs the suite (intended); Tier-1 secret never on fork runners
TM4 malicious keyed insider	correlation needs a public reference; Fork-A source is unpublished ? "not reasonably likely"	**residual:** conditional + revocable ? publish/leak the source later and the link can snap back (forward-looking re-test duty)
TM5 re-ID adversary	source unpublished + reset timeline ? no back-match against a public reference	? becomes live if a Fork-A source is ever published ? **and the committed per-frame trajectory IS a fingerprint** (de Montjoye 4-point re-ID), NOT just ~5 scalars (corrected §4c). The protection is the unpublished source, not minimization
TM6 accidental plaintext commit	required CI leak-scan rejects MD5/known-name/non-surrogate/plaintext-SOPS	if the required check is disabled, a leak lands

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6. Honest limits ? what a green suite does NOT prove

- Green proves only that the **deterministic post-trajectory transforms** are unchanged for the **frozen cases**. It does **not** cover: the WASB/TrackNet detector, `_probe_fps_width`/cv2 frame-seek, the AI/provider handoff, chunking/re-encode, the ffmpeg stitch, DB persistence/migrations, or the API filter. *A refactor that breaks decoding or the detector passes these gates green.* The owner-box `regression.py` DB run remains the only full-chain check.
- Characterization = **behaviour-locking, not correctness** ? it freezes current behaviour *including current bugs*. "Passing" = "behaviour unchanged", never "behaviour correct". This caveat must ride in the **test pass/fail message**, not only the docs.
- **Synthetic-tier floors = plumbing correctness, NOT field quality.** Never cite synthetic F1 in `QUALITY_ITERATIONS.md` / the paper as rally-detection performance (tag `kind: synthetic`).
- Tier-0 public coverage is **shuttle-only**; the person/audio fusion ?F1 and exact Gate-A values need Tier-1 (skipped on no-key clones).
- Data is **not "anonymous"** absolutely (EDPS v. SRB) ? relative, conditional,

revocable, forward-looking.
137 - The committed Tier-0 stream is the **de-attributed per-frame `trajectory.csv`**, NOT ~5 scalars ? a per-frame trajectory is a de Montjoye fingerprint; the only protection is the **unpublished source** (§4c ?, §5 TM5).
138 - **Time-origin reset is metric-invariant within `_FLOAT_TOL`, not byte-exact for every offset.** The genuine (raw) reset-vs-no-reset score delta is at the float-error floor (~3.3e-7); it is **bit-identical for whole-second offsets**, but because 6dp-rounding is not additive (`round6(a)-round6(b) ? round6(a-b)`) the *serialized* 6dp boundary error can differ by one 6dp ULP (1e-6) on a sub-second offset ? the serialization grid, not a metric divergence. (Audit fix #2 rounds `t0`+labels to 6dp before the shift to keep the residual ? the grid step.)
139 - Encryption gives **no secrecy from a running contributor**; protobuf gives **no secrecy at all**.
140 - De-attribution does **not** cure provenance/licensing ? it sits *behind* the per-record provenance gate + counsel sign-off.
141 - `--refresh-snapshot` can rubber-stamp a real regression; mitigation is **review of the visible diff** + a separate, deliberate floor-baseline update. **A recorded `--reason` is REQUIRED** (the CLI errors without one ? audit fix #6).
142
143 ---
144
145 ## 7. Deliverables & progress tracker ? **source of truth**
146
147 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. One **small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.
148
149 | ID | Deliverable | Depends on | Counsel-gated? | Status | PR |
150 |----|-------------|-----------|----------------|--------|----|
151 | **P0** | **PII leak hygiene (partial ? paths + scan).** Scrubbed personal `C:\Users\?` paths from code docstrings (`eval_partial_labels.py`, `gemini_refine.py`, `README.md`) + added `tools/leak_scan.py` (32-hex MD5 + personal-path **structural** guard, optional gitignored `.leakscan_private` name-list) + a CI `leak-scan` job + tests. ? The **functional video-name rename** (`eval_baselines/*.json` keys / `rally_seg_eval.MULTICOURT_CLIPS` / quality-log names) is **DEFERRED** per the owner's safe-scope decision (2026-06-16) ? it's coupled to the gitignored `output/human_lovo_manifest.json` + data files + the training floor and collides with the active quality track; run the private-list scan + a coordinated rename before open-sourcing. Git-history rewrite (PR #77) also deferred. | ? | No | ? | _this PR_ |
152 | **M1** | **Fixture framework + synthetic Tier-0 + shared replay helper** (also delivered M2-core + M3 ? see below). `backend/eval/golden_fixtures.py` (`replay_fixture`, schema dispatch, `Path(__file__)`-relative) + 3 **tool-generated** synthetic fixtures + manifest + README + `.gitattributes` LF pin + `tests/test_golden_fixtures.py`. CONTROL-only (no numpy execution on the replay path); cross-OS verified. | ? | No (synthetic) | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
153 | **M2** | **Freeze/refresh extractor.** Core (`freeze_synthetic` / `refresh_snapshot` / `--check` CLI + idempotence test) shipped in M1 (#189); the `--license`/`--minors-free`/`--owned` **refusal gate** shipped with P1 (this PR). | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) · _this PR_ |
154 | **M3** | **Characterization gate** ? `tests/test_golden_fixtures.py`: parse-compared (exact-int / approx-float 1e-6), **CONTROL-only**, positive no-person/audio assertion, perturbation + idempotence. **Shipped in M1 (#189).** (Windowing/label fingerprint guard + row-count cap ? fold into M4.) | M1 | No | ? | [#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189) |
155 | **MX** | **Cross-OS determinism guard (CI).** Lightweight `golden-crossos` CI job (ubuntu/windows/macos) running `--check` + the fixtures test ? locks the cross-OS guarantee (CONTROL replay is numpy-only, no GPU stack; a *separate fast job*, not full-suite×3). | M1 | No | ? | [#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191) |
156 | **M4** | **Quality-floor gate + first `regression_baseline.json`.** `tests/test_golden_quality_floor.py`; create `eval_baselines/regression_baseline.json` with **windowing fingerprint**; metrics-vs-base paired-delta; corpus-level (not per-slug) LOVO for runtime. | M1, M2 | No | ? | ? |
157 | **P1** | **De-attribution + minimization** ? `golden_fixtures.freeze_real_fixture()` (extends the M1 module, not a separate script): **opaque random `fx_<hex>` surrogate slug**, metadata strip, metric-invariant **time-origin reset**, strict GT loader, **positive no-person/audio/MD5/source-path assertion**, manifest non-clobber; `--freeze-real` CLI with the **refusal gate** (exit 2 unless owned + minors-free + license). Tooling-only, 12 synthetic tests; **no real data committed**. | M1 | No (tooling) | ? | _this PR_ |

P2	**Owner-accepted (researched-risk), 2026-06-16.** Owner accepted the researched IP/privacy analysis (§3) to proceed **under its conditions** (owned/Fork-A source · minors excluded · biometric/person streams excluded · de-attributed). Owner's informed risk acceptance ? **NOT a retained-counsel opinion**; optional future counsel confirmation per §8.	P1	Owner-accepted	?	?
P3	**Tier-0 REAL subset ? SHIPPED.** Committed **6 curated, trimmed (?12k-frame ? few-min), de-attributed** real fixtures (`fx_r01`?`fx_r06`: single-court / multi-court×2 / doubles×2 / continuously-active blob; both 29.97 & 59.94 fps) via `freeze_real_fixture --max-frames` (opaque slug, time-origin reset, scenario-only labels, **NO** source name/path/video_id ? grep-verified + leak-scan clean). ~1.9 MB total; the characterization + cross-OS gates now run on real data too. Owner cleared ALL owned golden videos (owned + minors-free) + re-confirmed option (a) (2026-06-22). The surrogate?source map stays **off-git** (owner-held).	P1, P2	conditions §3 ?	?	_this PR_
O1	*(optional)* **Protobuf + deterministic serialization.** `rally.fixture.v1` proto3, codegen + `protoc`/`git diff --exit-code` guard, canonical writer, `.gitattributes` binary. Only if cross-OS byte-stable snapshots are wanted.	M1	No	?	?
O2	*(optional)* **Tier-1 key-gated path.** `scripts/fetch_golden.py` (OneDrive token, default) and/or SOPS+age; `_tier1_present()`; `RALLY_GOLDEN_DIR` wired; Tier-1 CI job gated on secret, never on fork runners.	M1	No	?	?
DOC	**Discoverability cross-links** ? added the `eval_baselines/fixtures/` row to `CODE_MAP §0` + a `golden_fixtures.py` entry to `CODE_MAP §2.3`; cross-linked `GOLDEN_DATA_SHARING.md` (committed subset ? private OneDrive corpus). README indexed in #186; memory `current-state` kept live. (EVALUATION/QUALITY_ITERATIONS cross-links **deferred** ? that doc surface is hot in the concurrent quality track; avoid collision.)	M1	No	?	_this PR_

Must-fix-before-any-public-commit (red-team), folded into the rows above: ? numeric-tolerance not byte-equality + multi-OS matrix + CONTROL-only (M3/D5/MX ?); ? scrub names in **all** files + history decision (P0); ? leak-scan a **required** check, hooks are convenience only (P0); ? person-optional loader + **positive** no-person assertion, don't trust the silent `0.0` default (P1 ?); ? random surrogate id not HMAC-over-roster (P1 ?); ? quantize/drop fps/frame_width (P1 ? fps/fw retained as scale constants; quantize is a P3 option); ? document Tier-0 omits the fusion path (DOC); ? counsel + Fork-A + minors/biometric exclusion as hard preconditions (P2 owner-accepted; enforced in code by P1's refusal gate ?).

Progress log (quantitative ? coverage must hold/improve, CI incl. the 3-OS cross-OS job green)

PR	Deliverable(s)	Suite	Coverage	Notes
[#189](https://github.com/avidullu/badminton-highlight-indexer/pull/189)	M1 + M2-core + M3	746 pass / 7 skip	80.41%	synthetic Tier-0 + CONTROL replay + characterization; cross-OS proven (CI Linux replayed Windows-frozen fixtures)
[#191](https://github.com/avidullu/badminton-highlight-indexer/pull/191)	MX + tracker/legal reconcile	full suite green	80.41%	`golden-crossos` CI job green on ubuntu+windows+macos ? cross-OS **enforced**; legal ? owner-accepted
this PR	P1 + M2 refusal gate	759 pass / 7 skip	**80.49%** (+0.08)	`freeze_real_fixture` + refusal gate; **+13 tests**; tooling-only, no real data committed
this PR (audit-hardening)	adversarial-audit fixes #1?#17	784 pass / 7 skip	**80.71%** (+0.22)	strict-slug `label` + dropped `source_note` (#1); 6dp-rounded reset within `_FLOAT_TOL` (#2); temp-dir-then-promote + negative-label refusal (#3); exact-membership source-path scan (#4); key-set-equality in `compare_expected` (#5); `--reason` mandatory (#6); independent schema guard (#7); `max_win` recorded-not-applied note (#8); numpy-execution wording (#9); +failure-semantics & by-kind tests (#10/#11); `test_golden_real_fixtures.py` added to `golden-crossos` CI (#12); doc reconciled to shipped reality incl. per-frame-CSV honesty + risk re-confirm (#13?#17). Synthetic `--check` byte-identical (unchanged).
this PR (P0 safe scope)	personal-path scrub + leak-scan guard	1365 pass / 7 skip	85.27%	scrubbed `C:\Users\?` paths from 3 code docstrings (`eval_partial_labels.py` / `gemini_refine.py` / `README.md`); `tools/leak_scan.py` (32-hex MD5 + personal-path structural guard + gitignored `.leakscan_private` name-list) + a CI `leak-scan` job + 10 tests. Functional video-name rename DEFERRED (owner safe-scope).
this PR (P3 real fixtures)	6 curated trimmed de-attributed real Tier-0 fixtures + `--max-frames` trim	1390 pass / 7 skip	85.27%	`fx_r01`?`fx_r06` (~1.9 MB; single/multi-court/doubles/blob, both fps) frozen via `freeze_real_fixture --max-frames=12000`; de-attribution grep-verified + leak-scan clean; +trim test; owner re-confirmed option (a). F1 0.0?0.59 = real CONTROL-path (behaviour-locking, **NOT** field quality).

---

8. Open owner / counsel questions

Counsel questions ? documented basis (the owner accepted the researched risk on 2026-06-16; see P2). These remain the record, and a retained-counsel confirmation is optional/recommended before scaling beyond a small owned, minors-free, de-attributed subset:
1. Does the EU/UK **sui-generis database right** subsist in our self-derived shuttle/rally streams on the *Sportradar* "obtained vs created" reasoning, and does redistributing a de-attributed Tier-0 subset cause *CV-Online* harm?
2. Post *Thomson Reuters v. Ross* + USCO-2025, is **US fair use for the extraction step** defensible if the dataset could compete in a "data/analytics" market ? does publishing only numbers (never footage) matter?
3. India: are anti-scraping ToS enforceable; **IT Act s.43/s.66** exposure for any 3rd-party ingestion; does pending *ANI v. OpenAI* (~Apr 2026) move it? Confirm **DPDP s.9/Rule 10** and whether per-player tracking is prohibited s.9(3) monitoring of a minor.
4. Sign-off on the **final committed Tier-0 subset + provenance log**; is non-attributability-via-private-source a defensible posture to assert given its conditional/revocable nature?

Owner (strategic):
5. Publishing the Fork-A subset **forfeits the trade-secret moat** (Indian breach-of-confidence framing) ? weighed against community/benchmark/paper aims. Your call.
6. **Forward-looking commitment:** do not publish a highlight reel of the *same* source whose fixtures are committed (without re-clearing), and accept a periodic re-assessment as the public-video environment grows.
7. **Scope of first public commit:** synthetic-only (zero provenance risk, recommended) vs a counsel-cleared real subset.
8. **Tier-1 mechanism:** out-of-repo+token (recommended) vs SOPS+age in-repo. And: is protobuf (O1) wanted, or is canonical-JSON enough?

---

9. Definition of done

The project is **DONE** (flip `Status ? DONE`, archive) when:
- [ ] **P0, M1?M4, P1, DOC** are ? and merged.
- [ ] A clean `git clone` on **win + ubuntu + mac** runs `python run_all_tests.py` and the **Tier-0 gates pass green with no key / no video / no GPU / no DB**.
- [ ] The **CI leak-scan is a required status check** and passes (no MD5 / real-name / non-surrogate / plaintext-SOPS in the tree).
- [ ] Characterization + quality-floor gates are **parse-compared, CONTROL-only, windowing-fingerprinted**, and demonstrably catch a seeded behaviour change.
- [ ] `docs/CODE_MAP.md §0` + TEST MAP updated; this doc cross-linked; `NEXT_STEPS.md` + memory `current-state` point here.
- [ ] **Either** the public commit is synthetic-only, **or** (**P2 owner-accepted ? + P3**) for a real-data commit, under the §3 conditions enforced **in code** by P1's refusal gate.
- [ ] §6 honest-limits caveat rides in the test pass/fail messages, not only the docs.

Optional O1/O2 may remain ? without blocking DONE (they're earned, not assumed).

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10. References

Internal code anchors (verified): `backend/eval/{regression.py,fusion_golden.py,fusion_compare.py,ablation.py,experiment.py,metrics.py,windowing.py,distill_local.py,gt_loader.py,eval_stats.py}` · `backend/pipeline/detectors/{base.py,tracknet_runner.py,rally_gate.py}` · `backend/utils/hashing.py::compute_video_id` · `eval_baselines/{heuristic_lovo_n6.json,gemini_wrapper_2026-06-10.json,experiment_leaderboard.json}` · `tests/test_ablation.py::test_litmus_reproduces_gate_a` · `.github/workflows/ci.yml`.
Internal docs: `GOLDEN_DATA_SHARING.md` (#175), `EVALUATION.md`, `EXPERIMENT_HARNESS.md`, `QUALITY_ITERATIONS.md`, `MULTI_SIGNAL_FUSION_PLAN.md`, `CODE_MAP.md §0`.

Legal authorities: see §3 (full URLs in the research workflow artifact). *This section is researched general information, not legal advice.*

Changelog
- **2026-06-16** ? Initial draft synthesized from two multi-agent workflows (mechanics design + 2 critiques; legal memo IN/US/EU + SEALED-FIXTURES architecture + red-team). Verified the live name-leak and the BLAS-determinism cause against code. Status: DRAFT, owner-gated.

4. assistant (2026-07-06T20:14:04.638Z)

Let me read the remaining files in parallel.

5. user (2026-07-06T20:14:05.679Z)

Annotation & Evaluation

How to measure how well the pipeline detects rallies, using hand-annotated ground-truth timestamps. The harness scores predictions **already stored in the SQLite DB** ? it makes no API calls and is fully deterministic, so you can run it as often as you like for free.

---

1. What you annotate (and what it does / doesn't do)

You annotate, per video, the `[start, end)` time range of each **rally**. That's all the evaluation harness needs.

> **Important:** these rally timestamps measure and tune **rally segmentation**
> ? i.e. *where* the pipeline says rallies begin and end. They do **not** train
> the shuttle detector (TrackNet/WASB), which learns from per-frame shuttle
> `(x, y)` coordinate labels, a different and much heavier annotation. Use rally
> timestamps as your **golden evaluation set**; only invest in frame-level
> coordinate labels if evaluation shows *detection* (not segmentation) is the
> bottleneck. See `docs/TRACKNET_WSL_SETUP.md`.

2. Annotation file format (CSV)

One CSV per video, one rally per row:

```csv
start,end
00:01:12,00:01:39
102.5,131.0
```

- The `start,end` header is optional (auto-detected).
- Timestamps may be **decimal seconds** (`102.5`) or **colon-separated**
  `MM:SS` / `HH:MM:SS` (`00:01:12`). Forms can be mixed.
- Blank lines and `#` comments are ignored.
- Malformed rows (bad timestamp, `start >= end`) fail loudly rather than being
  silently skipped ? so labeling mistakes surface immediately.

Generate an empty template to fill in:

```bash
python run_eval.py --scaffold --annotations rallies.csv
```

3. Running the evaluation

First make sure the video has been processed so its detections are in the DB (the pipeline keys each video by its file MD5). Then:

```bash
# Identify the video by file (its MD5 is computed for you)...
python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv

# ...or by the stored video_id (MD5) directly:
python run_eval.py --video-id <md5> --annotations rallies.csv --stage both --iou 0.5
```

Useful flags:

Flag	Effect
`--stage {candidates,final,both}`	Which prediction stage(s) to score (default `both`).
`--detector <name>`	Only score candidates from one detector (e.g. `yolo_hybrid`, `tracknet_hybrid`).
`--iou <float>`	Match threshold (default `0.5`).
`--pr-curve`	Also report an IoU sweep (0.1?0.9).
`--show-failures`	List missed (FN) and spurious (FP) rallies with timestamps.
`--json <path>`	Write a machine-readable report.
`--db <path>`	Override the DB path (default: `output/<output.db_name>` from `config.json`).

4. Reading the output

```
================================================================
 Rally evaluation ? video 'demo'
 Ground-truth rallies: 3 | IoU threshold: 0.5
================================================================
 Stage         Pred   TP   FP   FN   Prec    Rec     F1   mIoU  dStart    dEnd
----------------------------------------------------------------
 candidates       3    3    0    0   1.00   1.00   1.00   1.00    0.0s    0.0s
 final            2    2    0    1   1.00   0.67   0.80   0.95    0.2s    0.5s
================================================================
 [final] missed rallies (false negatives): 1
   - 00:01:20?00:01:32
```

- **TP / FP / FN** ? true positives (matched), false positives (spurious
  predictions), false negatives (missed rallies).
- **Prec / Rec / F1** ? precision, recall, and their harmonic mean.
- **mIoU** ? mean temporal IoU over matched pairs (how tightly matched rallies
  overlap their ground truth).
- **dStart / dEnd** ? mean absolute boundary error in seconds (are rallies
  detected but trimmed in the wrong place?).

The key diagnostic: candidates vs. final

Scoring both stages tells you *where* to spend effort:

- **Candidates miss rallies the GT has** ? the **detector** is the bottleneck.
  Improve the shuttle model (e.g. fine-tune TrackNet/WASB with frame-level
  labels) or relax detection thresholds.
- **Candidates catch them but `final` is wrong/trimmed** ? the **segmentation /
  AI handoff** is the bottleneck. Tune thresholds (`velocity_threshold`,
  `dead_time_merge_gap`, `min_action_window_duration`) or the AI prompt ? no new
  labels needed.

In the example above, detection is perfect (candidates: recall 1.0) but the AI
handoff dropped one rally (final: recall 0.67) ? so the fix is in segmentation,
not detection.

5. How matching works

Predictions and ground truth are sets of `[start, end)` intervals. A prediction **matches** a ground-truth rally when their **temporal IoU** (overlap ÷ union) is at least the threshold. Matching is **greedy and one-to-one** ? highest-IoU pairs are claimed first, and each prediction/GT is used at most once, so a single detection can't be credited for two rallies. Unmatched predictions are false positives; unmatched ground-truth rallies are false negatives.

6. Code map

File	Role
`backend/eval/metrics.py`	Pure interval math: IoU, matching, P/R/F1, boundary error.
`backend/eval/annotations.py`	Load/validate the ground-truth CSV.
`backend/eval/harness.py`	Read predictions from the DB, score, format the report.
`run_eval.py`	CLI entry point.
`tests/test_eval_*.py`	Deterministic tests for all of the above.

7. Troubleshooting

Symptom	Cause & fix
`ERROR: database not found at output/...`	The harness only reads stored predictions ? it does **not** run the pipeline. **Process the video first** (e.g. `scripts/run_process_and_stitch.py`) so detections land in the DB, then re-run.
Every rally shows as a miss (FN), `Pred` is 0	This video hasn't been processed, or you used the wrong `video_id`. The pipeline keys videos by **file MD5** ? pass the *same* file via `--video`, or the exact stored `--video-id`.
`video_id` doesn't match although it's "the same" video	Re-encoding/trimming changes the bytes, hence the MD5. Use the exact file that was processed, or look up the stored id (it's printed during processing).
`candidates` stage is empty but `final` has data (or vice-versa)	Depends on which segmenter ran. `gemini` writes only `final` (no candidates); `yolo_hybrid`/`tracknet_hybrid` write both. Use `--stage final` for the gemini segmenter, and `--detector <name>` to scope candidates.
Recall looks low only at high `--iou`	Rallies *are* found but boundaries are loose ? check `dStart`/`dEnd`. Tune `stitching` padding or the segmentation thresholds rather than the detector.
`ValueError: start (...) >= end (...)` while loading annotations	A bad row in your CSV. Timestamps must be increasing per rally and use seconds or `HH:MM:SS`.

Run `python run_eval.py -h` for a quick reminder of this prerequisite and all flags.

6. user (2026-07-06T20:14:06.777Z)

Data in GCS ? the corpus source-of-truth map

> **Scope:** this map covers the **training corpus only** (`gs://khelsutra-rally-corpus`). **User-uploaded
> videos are a SEPARATE, already-shipped surface** ? `POST /api/assets` + the per-request `medium_uri` +
> `GET /api/capabilities.storage` (see `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`) ? and must stay physically
> isolated from the corpus bucket (the **D1** compliance boundary: consent / no-minors / not-training holds at
> the storage layer). The deploy layer's bucket names now live in one place (`deploy/gcp/buckets.env`).

Status: LIVE / in-progress (migration not complete). **Owner-gated** decisions in §7.

Last inventoried: 2026-06-22 against `gs://khelsutra-rally-corpus` (§5 gap table reconciled 2026-07-02 to the post-Phase-1 state ? it had kept showing the pre-Phase-1 gaps,

contradicting §7; Roora pilot non-video archive, Roora distilled golden-label fixtures, and six owned-source MD5s reconciled 2026-07-02).
Goal (owner, 2026-06-22): run **all** training + inference from the cloud with **all data in GCS**,
so the local Windows box (and the intermittently-mounted external drive) is **no longer a blocker**.

> **? Other sessions: this is where the data lives.** Pull from `gs://khelsutra-rally-corpus`, not
> from anyone's `C:` / `F:` / Google Drive. If something you need isn't in the bucket yet, it's in the
> **gap table (§5)** ? flag it / migrate it, don't hard-code a local path.

This doc is the **data map**. The *machine* runbooks are [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md)
(spin up a box) and [`deploy/gcp/README.md`](../deploy/gcp/README.md) (provision GPU + train/infer).
The cloud engine itself was built by the (archived) DECOUPLED_COMPUTE project
(`docs/archives/past_projects/DECOUPLED_COMPUTE/`); this doc does **not** redesign it ? it records
where the bytes are and *what's still missing*.

---

1. The bucket

Bucket	**`gs://khelsutra-rally-corpus`** (single corpus bucket)
Project	`gen-lang-client-0306026826` (billing *Khelsutra Billing*)
Region	`asia-south1` (Mumbai), UBLA, co-located with the GPU VMs
Auth	SA `rally-worker@gen-lang-client-0306026826.iam.gserviceaccount.com` (`roles/storage.objectAdmin`); key at `~/.gcp/rally-worker-sa.json` (local only, **never committed**). Interactive: `gcloud auth login`.
Size today	~2.06 GB (mostly past dry-run reels/outputs ? **not** the full corpus)

The engine is **backend-agnostic by URI** (`backend/storage/ref.py`): `gcs://` (or `gs://`),
`s3://`, `gdrive://`, or a bare local path. Pick the backend with **config + creds, no code change**.

---

2. Canonical layout (the target contract)

The cloud worker's input/output contract (`backend/worker/__main__.py`; lineage:
`DECOUPLED_COMPUTE/04`):

Prefix	Holds	Who reads/writes it
`videos/`	source videos / proxies (`<clip>.mp4`)	**input** to `worker infer` & `worker train`
`labels/`	golden rally labels (`<clip>.rallies.csv`)	**input** ground truth to `worker train` + all eval
`trajectories/`	cached WASB trajectory CSVs	? **eval/litmus input** (GPU-free), consumed via `manifests/golden.json`. **NOT** read by `worker train` (it synths or re-extracts)
`weights/<version>/`	trained Gen-0 bundles (`weights.json` + a run `manifest.json`)	**output** of `worker train`
`models/`	upstream pretrained weights (`wasb_badminton_best.pth.tar`)	**staged to a local path** (`$WASB_WEIGHTS` / `~/models/?`) for the detector ? not read from `gs://` in place
`outputs/<video_id>/`	per-video inference outputs (intervals, etc.)	**output** of `worker infer`
`highlights/`, `promo/`	rendered reels + promo artifacts	output / demo

`archive/roora-pilot-2026-06/`	Roora pilot non-video deliverables, review notes, sidecar CSVs, trajectory exports, and a SHA-256 manifest	**archive only**; raw videos remain owner-gated

> **Note on the manifest.** `worker train` does **not** read a pre-staged corpus `manifest.json` ? it
> *builds* one on-box at train time from `labels/` (+`videos/`), extracting trajectories **synthetically
> by default** (`--trajectory-source synth`, CPU-mock) or for real via opt-in `--trajectory-source
> detector` (needs `videos/`). A `manifest.json` only appears as an **output** inside each
> `weights/<version>/` bundle. The **eval/litmus** path is what wants a stable, path-resolvable manifest
> (§7.3) ? and `manifests/golden.json` is currently consumed by **no code yet** (eval is local-FS only;
> see Known blockers).

---

3. What's actually in the bucket today (2026-06-22)

> **Phase-1 landed 2026-06-22:** added `trajectories/` (15), 15 canonical `labels/<date>_<name>.rallies.csv`
> (= original-6 + `GX010137`/`GX010141` + 7 new; the 7 old `_proxy`/bare dupes were deleted), and
> `manifests/golden.json`. The table below is the **pre-Phase-1** snapshot; the remaining gap is **videos**
> (14 of 15) ? see §6/§7b/§8.

Prefix	Present	Drift / notes
`labels/`	7 files: `GX010128`, `mahadevpura-1`, + `Badminton BXH 2_proxy`, `Boxhill Doubles_proxy`, `GX020094_proxy`, `GX030094_proxy`, `mahadevpura-2_proxy`	? **inconsistent naming** (`_proxy` on some, not others); ? **missing** the original-6 labels + `GX010137`, `GX010141`
`videos/`	3: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4`	? corpus videos/proxies almost entirely absent
`trajectories/`	?	? **does not exist** (every trajectory is local-only)
`models/`	`wasb_badminton_best.pth.tar`	? (? WASB-C3 weight-provenance still an open legal item before any external share)
`weights/`	`0.1.0-l4/`	? first real L4 train bundle
`outputs/`	one `<video_id>/`, `parity/`	? dry-run outputs
`highlights/`, `promo/`	reels + promo	?

Drift summary: the bucket has extra prefixes (`highlights/`, `promo/`, `models/` alongside `weights/`) and **inconsistent `labels/` naming**, and is **missing the bulk of the corpus inputs** (`videos/`, all `trajectories/`, most `labels/`). It reflects past *dry runs*, not a complete corpus.

Roora pilot archive addendum (2026-07-02): the non-video pilot record is now backed up under `gs://khelsutra-rally-corpus/archive/roora-pilot-2026-06/`. The prefix contains 740 objects total: 739 manifest entries (5,731,641 bytes) plus `manifest.json`. Verification on 2026-07-02 found **0 video objects** under the prefix. The manifest schema is `khelsutra.roora_pilot_archive.v1` and records relative local source paths, destination `gs://` URIs, byte sizes, and SHA-256 hashes. Raw pilot videos under `Annotation Videos/` were deliberately excluded. The distilled human-corrected Roora goldens are also git-versioned in `sports-data-collector/data/golden_labels/` as of `sports-data-collector#56`.

---

4. How the cloud consumes it

Provisioning + the verified commands live in [`deploy/gcp/README.md`](../deploy/gcp/README.md) §3?§5.
The data-facing entry points (`backend/worker/__main__.py`):

```bash
# config (no secrets in config; auth via GOOGLE_APPLICATION_CREDENTIALS):
#   { "storage": { "backend": "local", "gcs": { "project": "gen-lang-client-0306026826" } } }

# INFERENCE on one cloud video -> outputs/<video_id>/ in the bucket
python -m backend.worker infer \
  --video-uri gcs://khelsutra-rally-corpus/videos/<clip>.mp4 \
  --out-uri   gcs://khelsutra-rally-corpus --segmenter wasb_hybrid --config config.json

# TRAIN from the bucket (reads labels/ [+ videos/], extracts trajectories, builds manifest on-box,
# runs training.gen0.harness, pushes weights/<version>/):
RALLY_DETECTOR_IMPL=native python -m backend.worker train \
  --corpus-uri gcs://khelsutra-rally-corpus --out-uri gcs://khelsutra-rally-corpus \
  --version <v> --trajectory-source detector --segmenter wasb_hybrid --config config.json
```

A bare/relative path resolves under a configured `input_root`/`input_backend`
(`backend/storage/ref.py::resolve_input_uri`), so workflows can be written bucket-relative.

---

5. The gap ? what's still local-only (why the box is still a blocker)

> **Reconciled 2026-07-02.** Phase-1 corpus *metadata* (trajectories + labels + manifest) **landed in
> the bucket 2026-06-22** ? see §6 (Phase 1) and §7 decisions #2/#3 (LOCKED/done). This table previously
> still showed those as ? and contradicted §7/§7b. Post-Phase-1 the remaining gaps are the **videos**
> (Phase 2) and the **Phase-3 eval-on-GCS plumbing** (§6 Phase 3 ? the eval drivers still read local
> paths with plain `open()`, so bucket-resident metadata isn't consumed by eval yet).

Data	Where it is now	Size	In bucket?	Needed for
WASB trajectory CSVs	bucket **+** local working copy (`?\Annotation Setup\?\Trajectories\`)	**17.5 MB**	? canonical `trajectories/<name>.csv` (2026-06-22, §7.2)	eval/litmus (cached). *(No `cached` train source exists ? train synths or detector-extracts.)*
golden manifest	`gs://?/manifests/golden.json` (§7.3); the **local** litmus still reads `output/human_lovo_manifest.json`	4.5 KB	? (2026-06-22, §7.3) ? flipping the litmus to read it is Phase 3	eval corpus definition
original-6 labels	bucket **+** F: archive	~12 KB	? among the 15 canonical `labels/<date>_<name>.rallies.csv` (2026-06-22, §7.1)	train + eval
7 new-clip labels	bucket **+** repo `output/*.rallies.csv` (gitignored) **+** F: archive	~12 KB	? ditto (naming canonicalized 2026-06-22; the stale `_proxy`/bare dupes MD5-verified + deleted, §7)	train + eval
source videos / proxies	F: archive + Google Drive golden folder; six owned-source MD5s in `sports-data-collector/data/golden_source_videos.json`	**100s of GB**	? mostly not migrated ? **the remaining large gap** (Phase 2; see §6)	infer + train (real WASB extraction)

|
| Roora pilot non-video record | `Annotation Setup\Archive\Roora Pilot Review - 6 June 2026` + root sidecar CSVs + `Trajectories\roora`; distilled goldens also in `sports-data-collector/data/golden_labels/` | **5.7 MB** | ? `archive/roora-pilot-2026-06/` (2026-07-02 manifest + 0 videos); small goldens git-versioned in `sports-data-collector#56` | vendor-pilot archive / loss prevention |
| Roora pilot raw videos | `Annotation Setup\Archive\Roora Pilot Review - 6 June 2026\Annotation Videos` | **4.8 GB** | ? owner-gated; not uploaded by the metadata archive pass | vendor-pilot provenance |

> Post-Phase-1 the corpus *metadata* is **in the bucket**; what still blocks GPU-free cloud eval is the
> **Phase-3 plumbing** (§6 ? the eval drivers can't fetch `gs://` yet). The **videos** are the only large
> migration left (Phase 2), and they already exist on **Google Drive** (the golden-corpus folder), so they
> can go **Drive ? bucket from a cloud box**, keeping the local machine out of the path entirely.

---

6. Migration plan (close the gap)

Phase 1 ? corpus metadata (?25 MB, do first, removes the eval blocker). ? DONE 2026-06-22
(see §7 decisions #1/#2/#3 + §7 cleanup note; the sketch below is kept as the historical plan). Stage all
trajectories, all labels, and a **bucket-relative eval manifest** into `gs://khelsutra-rally-corpus`.
Sketch:

```bash
# trajectories (decision-free; absent today)
gcloud storage cp "C:\?\Annotation Setup\Trajectories\*.csv" gs://khelsutra-rally-corpus/trajectories/
# labels (after the §7.1 naming decision) -> labels/<canonical-name>.rallies.csv
# eval manifest with gs:// (or bucket-relative) paths -> a stable manifest object (§7.3)
```

Phase 2 ? videos ? `videos/<name>.mp4` (bucket name = the eval `name`, consistent with
`trajectories/` + `labels/`). Sources (resolved 2026-06-22 ? see `scratch/copy_videos_to_gcs.ps1`):

Source	Clips	Size
Drive golden folder `?/Golden Label Videos Request - June 15/[Done] Video N/` (**proxies**)	`mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles` (7)	~13.5 GB
F: archive (**full MP4s** ? proxy-first before upload; owner-box root verified as `F:\[Khelsutra] GoPro Hero Black 12\GoPro Recorded Videos`)	`GX010137`, `GX010141`, + original-6 (`adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`, `testlarge_short`, `mahadevpura_singles`) (8)	~50 GB

Two ways to run it:
- **Decoupled (preferred ? no local box):** `rclone` on a cloud VM with a Google-Drive remote ?
  `rclone copy gdrive:"<?/[Done] Video N>/<proxy>.mp4" :gcs:khelsutra-rally-corpus/videos/<name>.mp4`
  (one-time owner step: `rclone config` ? `gdrive` remote OAuth). Or `gdown <file-id>` on the box for
  the shared folder. This is the path that actually removes the local box.
- **Interim (today, owner box ? streams Drive/F: ? local ? GCS):** `pwsh scratch/copy_videos_to_gcs.ps1
  -Execute` (dry-run without `-Execute`; `-Only <name>` for one; skip-existing). Uses

local bandwidth.

Priority: the 7 Drive **proxies** first (already 1080p, sufficient for detection). The 8 large F: full MP4s should be **proxied to 1080p first** (cheaper storage + faster cloud decode) before upload. ? Two pre-existing non-canonical video objects (`videos/GX030094_proxy.mp4`, `videos/mahadevpura-1.MP4`) are superseded once `videos/GX030094.mp4` / `videos/mahadevpura_1.mp4` land ? verify-dupe-then-delete (same as the label cleanup). Track per-clip in §8.

Phase 3 ? make eval/litmus read GCS (real code, NOT a config flip). The eval drivers (`served_gate_a.evaluate_video`, `calibrate_wasb.load_golden_gts`, `TrackNetRunner.parse_trajectory_csv`) open `spec["trajectory"]`/`spec["labels"]` with plain `open()` and gate on `os.path.isfile()` ? they have **no storage-layer fetch**, so a `gs://` manifest cannot be consumed today. Phase-3 must thread `storage.fetch`/`resolve_input_uri` through those paths first. Then: re-probe `fps`/`frame_width` from the bucket videos (don't trust the copied values ? a wrong fps silently mis-aligns every label), and **re-measure the served Gate-A litmus on the GCS 15-clip set as a [[launch-report-convention]] event** (dual-set RAW+golden no-regression, not a silent eval-set swap ? see PR #236).

---

7. Decisions

1. **`labels/` naming ? LOCKED (owner, 2026-06-22):** `labels/<YYYY-MM-DD>_<name>.rallies.csv`, where `<name>` is the eval/manifest clip name and `<YYYY-MM-DD>` is the label's **authoring date** = the *earliest* mtime across all known copies (C: working copy + F: archive ? robust to copy-resets; a restored C: copy shows today, the F: archive keeps the real date). Fixes the `_proxy` inconsistency and disambiguates recycled GoPro names by date. *(Future, when videos are in the bucket: optionally add a `video_id`/MD5 sidecar for the strongest keying ? deferred, needs hashing each video.)*
2. **Cache `trajectories/` ? YES, done 2026-06-22.** `trajectories/<name>.csv` cached so cloud/CI eval + the served Gate-A litmus run **GPU-free** (training can still regenerate via `--trajectory-source detector`).
3. **Eval manifest in GCS ? done 2026-06-22.** `gs://khelsutra-rally-corpus/manifests/golden.json` (15 clips, `gs://` trajectory + label paths). The **local** litmus keeps its local-path manifest; flipping the litmus to read the GCS manifest is Phase 3.
4. **Originals migration scope ? OPEN** ? proxies-only vs full 4K, and how much of the 15-clip golden set vs the wider archive. Source = Google Drive ? bucket from a cloud box (not the local box).
5. **Roora pilot non-video archive ? done 2026-07-02.** `archive/roora-pilot-2026-06/` backs up CVAT/vendor deliverables, feedback/review notes, root sidecar CSVs, and `Trajectories/roora` sidecars with SHA-256 hashes in `manifest.json`; the small human-corrected badminton/pickleball/ table-tennis goldens are git-versioned in `sports-data-collector#56`; raw videos stay excluded pending owner upload scope.

6. **Owned-source MD5 record ? done 2026-07-02.** The six original owned golden MP4s on `F:` were
   MD5-hashed and the collector registry paths were refreshed in `sports-data-collector#60`; the
   durable manifest is `sports-data-collector/data/golden_source_videos.json`. Uploading those large
   originals (or generated proxies) remains the Phase-2 owner-gated decision.

> Phase-1 was run via a dry-run planner (resolve every clip's trajectory + label, stamp the label's
> authoring date, build the `gs://` manifest) ? `gcloud storage cp` (planner kept in gitignored
> `scratch/`; promote it to `scripts/` if corpus-sync recurs).
> ? **Cleanup DONE 2026-06-22:** the 7 pre-existing inconsistent label objects (`*_proxy.rallies.csv`,
> bare `GX010128`/`mahadevpura-1`) were confirmed **byte-identical (MD5)** to the new canonical names
> (content also on C:/F:) and **deleted** ? `labels/` now holds exactly the 15 canonical objects.

> ? **gcloud gotcha (Phase-2):** `gcloud storage cp` treats `[...]` as a glob, so source paths under
> `[Done] Video N/` (Drive) or `[Khelsutra] ?` (F:) fail with "matched no objects." The owner-box
> script (`scratch/copy_videos_to_gcs.ps1`) stages such paths through a bracket-free temp file; the
> `rclone` cloud path has no such issue.

7b. Known blockers ? must-fix before the DoD (2026-06-22 adversarial review)

1. **? RESOLVED 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319))
   ? dated labels broke `worker train --trajectory-source detector`.** *History:* `cmd_train` derived the
   join key as `fname.replace(".rallies.csv","")` ? `2026-06-22_GX020094`, then looked up
   `video_by_stem.get(name)` against bare video stems (`GX020094`), so every clip was skipped and train
   aborted with `<2 videos`. **Fix:** the label?clip-name derivation is now the helper `_label_clip_name()`
   in [backend/worker/__main__.py](../backend/worker/__main__.py), which strips both the
   `.rallies.csv`/`.csv` suffix AND a leading `^\d{4}-\d{2}-\d{2}_` prefix, so a dated label and a legacy
   bare label both resolve to the bare `videos/<name>` stem (preserves the locked human-readable naming).
   Tested: parametrized helper cases + a dated-label?bare-video `cmd_train` join regression + per-label
   skip-branch coverage. *(`synth` train + all eval were unaffected ? they don't join videos.)*
2. **Phase-3 eval-on-GCS is net-new plumbing, not a flag** ? see §6 Phase-3. Eval is local-FS only.
3. **`manifests/golden.json` is consumed by zero code today** (train ignores it; eval can't fetch
   `gs://`) ? it's correct-but-dormant until blocker #2 lands. Add a tiny manifest-validator + a CI job
   that authenticates to GCS and runs the litmus against it, or the DoD ("CI reads only from `gs://`")
   stays unverifiable.
4. **Gating infra (owner-only, not done):** GPU `GPUS_ALL_REGIONS` quota is still **0** (no VM ever
   created ? `deploy/gcp/README` §2); cloud-box secrets (SA JSON + rotated `GEMINI_API_KEY` + WASB
   weights staged from `models/`) aren't set up persistently. No real cloud train/infer

runs until these.

5. **Reproducibility:** the migration + copy helpers live in gitignored `scratch/` (`migrate_golden_to_gcs.py`, `copy_videos_to_gcs.ps1`) ? promote to `scripts/` so Phase-2/re-syncs are repeatable from the repo, and add a referenced **proxy-generation** step for the 8 large F: originals.

8. Progress tracker

Item	Status
Bucket + SA + provisioning (deploy/gcp)	? built
Cloud worker `infer`/`train` by URI	? built (`backend/worker`)
This data-map doc	? 2026-06-22
Phase 1: trajectories ? `trajectories/` (15)	? 2026-06-22
Phase 1: labels reconciled ? `labels/<date>_<name>` (15)	? 2026-06-22
Phase 1: eval manifest ? `manifests/golden.json`	? 2026-06-22
Phase 1: remove 7 stale `_proxy`/bare label objects	? 2026-06-22 (MD5-verified dupes, deleted)
Blocker §7b#1: dated labels join in `worker train --trajectory-source detector`	? 2026-06-22 ([#319](https://github.com/Khelsutra/badminton-highlight-indexer/pull/319)) ? `_label_clip_name` strips the ISO-date prefix
Roora pilot non-video archive ? `archive/roora-pilot-2026-06/`	? 2026-07-02 ? 739 manifest entries + `manifest.json`; verified 740 objects and 0 video objects in the prefix
Roora pilot distilled golden labels	? 2026-07-02 (`sports-data-collector#56`) ? human-corrected badminton + pickleball/table-tennis goldens are git-versioned with tests
Six owned golden source MP4 hashes	? 2026-07-02 (`sports-data-collector#60`) ? MD5/path manifest committed; upload still owner-gated
Phase 2: corpus videos/proxies ? `videos/<name>.mp4`	? mapping + script done; **1/15** demo'd (`mahadevpura_1`); 14 remain ? run `rclone` on a cloud box (preferred) or the owner-box script
Phase 2: proxy the 8 large F: full-MP4 clips before upload	?
Phase 3: eval/litmus default to the GCS manifest	?

Definition of done: a fresh GPU box (or CI) can run inference, the eval suite, and Gen-0 training reading **only** from `gs://khelsutra-rally-corpus` ? no `C:` / `F:` / local-Drive path required.

Related
- [`SETUP_NEW_MACHINE.md`](SETUP_NEW_MACHINE.md) ? spin up a box (storage backends in §7).
- [`deploy/gcp/README.md`](../deploy/gcp/README.md) ? GCP GPU provisioning + train/infer (§5).
- [`VIDEO_LOCALITY_MODEL.md`](VIDEO_LOCALITY_MODEL.md) · [`PER_VIDEO_WORKSPACE.md`](PER_VIDEO_WORKSPACE.md) ? locality tiers + `outputs/<video_id>/`.
- `docs/archives/past_projects/DECOUPLED_COMPUTE/` ? the cloud-engine design lineage.

7. user (2026-07-06T20:14:07.059Z)

R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)

Status: **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked 2026-06-17; `backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive; tIoU stays the gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set regression gate.
A focused increment of the rally-quality project (RALLY_QUALITY_RESEARCH.md §7;
the day-1 validation that motivates it: REAL_FOOTAGE_VALIDATION_REPORT.md;

the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).

> **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is computed over
> **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set is survivorship-biased
> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to surface; that
> within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation method. Measured on the
> n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap", confirmed.

---

0. TL;DR
`tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-penalizes the
intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap matching is murky about
over-production. R2 replaces the headline with a **tolerance-match metric under strict 1:1 (Hungarian)
assignment**, reported as a **decomposition** ? `P@? / R@? / F1@?` + **boundary-error distributions split
into start/end** + an **over-segmentation guard** + a **?-sweep** ? not a single number.

Honest, slightly uncomfortable consequence (measured, see §3): under R2 the local detector's headline
number *drops* and the gap to Gemini *widens* vs what tIoU showed ? because R2 faithfully charges for the
1.5× over-production and the loose ends that tIoU's overlap matching partly absorbed. That is the point:
R2 splits "behind Gemini" into its two real causes ? **(a) over-production [precision]** and **(b) loose
rally-ends [the R4 lever]** ? and forgives only the inherent start fuzziness. It is a more honest ruler,
not a flattering one.

---

1. Why `tIoU@0.5` is the wrong ruler (grounded in the n=2 ground-truth clips)
1. **Over-penalizes the START.** For a 5.7 s rally, `tIoU ? 0.5` tolerates only ~±1.9 s of boundary shift
   (`? = D·(1??)/(1+?)`). The service window (stance ? racquet contact) alone is ~1?1.5 s ? even **Gemini's**
   median |?start| is 0.98?1.46 s. So tIoU spends most of its budget on a boundary that is *intrinsically*
   undefined to ~1 s. (Owner directive: don't strictly align starts; tolerate ~±2 s.)
2. **Count parity is a deceptive proxy.** mahadevpura-2 hit 1.03× of Gemini's count yet F1@0.5 = 0.150 ?
   right count, wrong boundaries. A count- or overlap-based view hides this.
3. **It flips config rankings on noise.** The hard-cap was golden-neutral at n=6 (+0.006 n.s.) but "best" at
   n=7 once a continuously-active clip entered ? a metric artifact at the 0.5 cliff, not a quality change.
4. **The deficits we must see are start?end-symmetric.** Local's |?start| ? Gemini's; the **entire** gap is
   the rally-**END** (|?end| 2.1?3.1 s vs Gemini ~0.8?1.2 s). A single blended IoU cannot show that ? and
   directing R4 needs exactly that decomposition.

---

2. The metric

2.1 Match rule (asymmetric tolerance ? forgive the start, expose the end)
A predicted window matches a golden rally iff **`|?start| ? ?_start` AND `|?end| ? ?_end`**.
- **`?_start` generous (~2.0 s)** ? absorbs the service-window/annotator start fuzziness (the part that is *not* a model deficit).
- **`?_end` tighter (~1.5 s, but IAA-calibrated ? see §2.4)** ? the rally-end IS a sharp physical event (shuttle down/net/out), so a loose end *is* a real deficit we want the metric to expose, not forgive.

> A *symmetric* ±2 s tolerance is tempting (the owner's stated bar) but the preview (§3) shows it is **not**
> automatically more lenient than tIoU 0.5 and it blurs exactly the end-deficit we need to see. Asymmetry is
> the point: relax the dimension that's intrinsically fuzzy (start), keep honest the one that's a real event (end).

2.2 Assignment ? strict 1:1 (Hungarian), never overlap/greedy
Match references to predictions by **global Hungarian assignment** (max total matches under the tolerance gate), one-to-one. This is what makes **over-production cost precision directly**: `P@? = matched / n_pred` crashes when the detector emits 75 windows for 52 rallies. (tIoU's overlap matching let extras "match something"; the W1 over-segmentation pendulum was barely costed.)

2.3 Report the DECOMPOSITION, never one number
1. **`P@?`, `R@?`, `F1@?`** ? precision (over-production), recall (did we find the rally within tolerance), F1.
2. **Boundary-error distributions** ? median **+ IQR** of signed *and* absolute `?start`, `?end` over matched pairs, **split**. This is the metric that says *"start solved, end open"* and aims R4.
3. **Over-seg guard (mandatory):** `split_count` (preds covering one ref ? W1 over-seg) and `merge_count` (refs under one pred ? the baseline mega-window). The *original* rule was strict per-video: **a promotion may not increase either on ANY video.** **REFINED to a NET guard, 2026-06-17 (R1, owner-approved)** ? the strict rule is correct for an *incremental* cue but **mis-fires on a structural mega-window?bounded change** (splitting a mega-window necessarily lifts `split_count` from ~0, and the raw merge *count* is non-monotonic: one window hiding 40 rallies is `count=1, rate=1.0`; bounding it into pieces that each glue 2 neighbours is `count=9, rate=0.5` ? FEWER rallies hidden). The net guard scores over-seg on the **normalized rates** as one weighted cost per video (`w_merge·merge_rate + w_split·split_rate`, merges weighted higher ? a merge hides a rally = recall loss, a split only fragments one), and **passes iff (a) the corpus-mean net cost does not rise AND (b) no video's `merge_rate` regresses.** Implemented as `tolerance_metrics.over_seg_guard` (it always reports the strict verdict too; revert = read `strict_passes`). See QUALITY_ITERATIONS.md "R1" for the measured verdict (milestone: mean `merge_rate` 0.694?0.150, net PASS, strict BLOCK).
4. **?-sweep** (`? ? {1.5, 2.0, 2.5, 3.0}`, symmetric for the trend; asymmetric pair for the headline) ? because the convention/IAA noise is large, a single ? misleads (§3). Report the curve.
5. **`tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-ablation backbone comparable) ? **not the gate.**

2.4 Choosing ? ? it MUST be calibrated to inter-annotator agreement
The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half the rallies the golden END differs from Gemini's by >1.5 s. That is plausibly **label/convention noise, not model error**: we'd be measuring how two humans (or human-vs-Gemini) disagree on "when did the rally end." **Therefore `?_end` must be ? the inter-annotator end-agreement**, which we do not yet know. **Action:** a small **IAA study** ? a 2nd labeler re-labels 2?3 already-golden videos; set `?_start`/`?_end` to ~the 90th-percentile pairwise human |?start|/|?end|. Until then **freeze provisional ?_start=2.0, ?_end=1.5 and always report the sweep**, with the `? = D·(1??)/(1+?)` derivation recorded so the choice is auditable.

---

3. Preview on the two ground-truth clips (greedy-1:1 approximation; Hungarian in the impl)

Symmetric-? sweep, F1@? (local-best = the per-clip best config: mp2 W1+W2+hardcap, GX010128 W1+W2):

clip / method	tolF1@1.5	@2.0	@2.5	@3.0	(tIoU-F1@0.5)
mahadevpura-2 / **Gemini**	0.203	0.354	0.506	**0.557**	0.557
mahadevpura-2 / local	0.111	0.148	0.222	0.278	0.296
GX010128 / **Gemini**	0.615	0.769	0.813	**0.857**	0.813
GX010128 / local	0.189	0.315	0.346	0.362	0.457

Reading the preview (all honest):
- **`tolF1@~3.0 ? tIoU-F1@0.5`** ? the sweep is a smooth, interpretable knob that *contains* tIoU as a point; good (continuity with the frozen baseline).
- **The local?Gemini gap WIDENS under the tighter, 1:1 tolerance** (GX010128 ratio 0.56 at tIoU0.5 ? **0.41** at ?2.0). R2 charges local for the 1.5× over-production (precision) and the loose ends ? exactly the deficits tIoU under-counted. **This is the metric working, not local regressing.**
- **Even Gemini drops at small ?** (mp2 0.557?0.203 from ?3.0?1.5) ? strong evidence the **golden END convention carries >1.5 s of noise** ? §2.4's IAA calibration is a prerequisite, not a nicety.
- Boundary-error medians (the robust diagnostic, from the validation report): local |?start| 0.78?1.74 s (? Gemini 0.98?1.46) vs local |?end| 2.12?3.12 s (? Gemini 0.76?1.19) ? **the headline R4 signal.**

---

4. Promotion / "trustworthiness" rule (small-n)
A candidate is promotable only if **all** hold:
1. **Paired per-video sign-test sweep on F1@?** is near-unanimous: **6/0 @ n=6 (p=0.031)** or **7/0 @ n=7 (p=0.016)** (8/0 @ n=8, p=0.008); 5/1 or 6/1 is NOT significant. Run on the **growing-golden** set.
2. **No over-seg regression** ? the **net** over-seg guard passes (§2.3.3): the corpus-mean weighted merge/split-*rate* cost does not rise AND no video's `merge_rate` regresses. (Was the strict per-video count rule; refined to net on 2026-06-17 ? see §2.3.3.)
3. **Effect size + bootstrap CI** reported as an honesty band (expect it to span 0 at this n; use it to separate "real" from "directionally-clean-but-negligible," not as the pass/fail).
4. **Stratify, don't blend** ? report the continuously-active blob clip (mahadevpura-1)

as its own stratum (its near-degenerate scores can swing a small-n mean).

5. **No regression on EITHER eval set** (raw + growing-golden) ? the dual-set discipline.

---

5. Implementation plan (pure, offline, additive ? zero production risk)

- **`backend/eval/tolerance_metrics.py`** (pure, GPU-free, unit-testable):
  - `match_1to1(preds, gts, ?_start, ?_end) -> (matches, P, R, F1)` ? Hungarian via `scipy.optimize.linear_sum_assignment` if available, else a pure O(n³) fallback (corpus n is tiny).
  - `boundary_error_report(preds, gts) -> {?start, ?end: signed/abs median+IQR}` (computed over unconditional best-overlap matches internally ? see the refinement note above).
  - `over_seg_report(preds, gts) -> {split_count, merge_count}` (reuse the bipartite logic from `segmentation_metrics.merge_split_report`).
- **Wire into `backend/eval/rally_seg_eval.py`** ? emit the R2 block **alongside** tIoU (don't remove tIoU); add `--tau-start/--tau-end` (both landed; the ?-sweep helper `tolerance_metrics.tolerance_sweep` exists but is not yet surfaced as a `rally_seg_eval` CLI flag). Surface in the experiment leaderboard.
- **`run_regression.py`** ? add the dual-set R2 aggregate + the over-seg-guard gate. *(Status: the guard primitive landed as `tolerance_metrics.over_seg_guard` and is wired into `rally_seg_eval.compare()`/`--vs` ? the natural home, since `rally_seg_eval` loads the golden manifest + trajectories. The dual-set `run_regression.py` aggregate is still a separate future item.)*
- **Tests** ? synthetic intervals: exact-match, start-fuzzy-within-?, end-loose-beyond-?, over-production (precision crash), the Hungarian-vs-greedy tie case, empty inputs.
- **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU (augment). The *gate* switches to R2 only after a dedicated **ablation experiment** on the golden corpus shows R2 ranks candidates at least as sensibly as tIoU ? so the promotion bar never changes silently mid-flight. When tIoU is retired as the gate it is **deprecated with documentation, not deleted** (kept as the secondary trend-line + the `QUALITY_ITERATIONS.md` paper-ablation backbone, with a dated note recording the ablation evidence for the switch). Owner decision, 2026-06-17 (§7).

---

6. Definition of done

- `tolerance_metrics.py` + tests merged (default-additive); `rally_seg_eval` reports the R2 decomposition next to tIoU (the ?-sweep helper exists in `tolerance_metrics` but is not yet wired into the `rally_seg_eval` report); the corpus (n?8) re-scored under R2 and recorded in QUALITY_ITERATIONS.
- The IAA study (§2.4) scoped (it can land after R2's code; R2 ships with provisional ? + the sweep).
- `run_regression.py` carries the dual-set R2 gate + the per-video over-seg guard ? but R2 **augments** tIoU until the ablation experiment (§7) validates it as the gate; only then does the gate switch.
- tIoU/mIoU retained as a secondary trend-line; when retired as the gate, **deprecated-with-docs (kept for posterity)** per the §7 owner decision ? a dated note records why and links the ablation evidence.

---

7. Decisions
Locked (recommended, design-level): Hungarian 1:1 (not greedy/overlap); report the decomposition + sweep, never one number; tIoU demoted to secondary; R2 default-additive until validated; boundary-error split start/end is the R4-aiming diagnostic.

Locked (owner review, 2026-06-17):
- **?_end asymmetric (1.5 s) vs ?_start (2.0 s) ? ADOPTED.** Asymmetry is the point: forgive the intrinsically fuzzy start, keep the rally-end honest. Your "±2 s, not more" bar maps to `?_start=2.0`; `?_end` stays the tighter, IAA-calibrated knob (§2.4).
- **IAA study after R2 code ? ADOPTED.** R2 ships now with provisional `?_start=2.0 / ?_end=1.5` + the full sweep; the IAA study lands next to fix ?. R2's code is not blocked on it.
- **Gate: AUGMENT now, then ablation-gate the eventual replacement, then deprecate-with-docs ? ADOPTED.** R2 is reported **alongside** tIoU (augment) until validated. tIoU is **not** swapped out silently: before R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden corpus showing R2 ranks candidates at least as sensibly as tIoU (and exposes the deficits tIoU hid). When tIoU is finally retired as the gate, it is **formally deprecated with documentation ? kept for posterity, never deleted**: it remains a secondary trend-line and the paper-ablation backbone (`QUALITY_ITERATIONS.md`), with a dated note recording *why* it was superseded and the ablation evidence behind the switch. (Consistent with the repo's "retain weak/superseded signals; numbers inform, foundations decide" norm.)

8. assistant (2026-07-06T20:14:09.574Z)

Now let me read the three files in `docs\data_pipeline\`. Let me first verify their exact paths.

9. user (2026-07-06T20:14:10.087Z)

docs\data_pipeline\GOLDEN_DATA_SHARING.md
docs\data_pipeline\GOLDEN_SET_IMPLEMENTATION.md
docs\data_pipeline\GOLDEN_SET_PHASE1_VIDEOS.md
docs\data_pipeline\README.md
docs\data_pipeline\VIDEO_RECORDING_GUIDELINES.md
docs\data_pipeline\GOLDEN_VIDEOS.md

10. assistant (2026-07-06T20:14:11.973Z)

All three requested files exist. Let me read them fully.

11. user (2026-07-06T20:14:13.479Z)

Golden-labeled video registry

Purpose. The single source of truth for every **golden** (human rally-boundary-labeled) video ? what it is, where the source proxy lives, how many rallies, and whether it's ingested into the eval corpus. Use this to **isolate the golden videos into a dedicated location** and to keep the growing golden set auditable. Updated whenever a new video is labeled + ingested.

> **This is the GROWING GOLDEN-finalized set** in the dual-eval-set discipline (see
> [QUALITY_ITERATIONS.md](../QUALITY_ITERATIONS.md), the 2026-06-16 capstone + `eval-dual-set-regression`):
> a **RAW eval set** (currently-bad videos) + this **golden-finalized eval set** (promoted when at-par) +
> a separate **golden TRAINING set**. Every quality launch must not regress on either eval set.
> Privacy-safe, video-free *shipping* of these = [GOLDEN_REGRESSION_FIXTURES.md](../GOLDEN_REGRESSION_FIXTURES.md).

Contracts / conventions
- **Labels:** rally-annotator CSV (`rally_number,start_time,end_time,ending_reason,sport,shots_count`,
  decimal seconds) ? ingested by `backend/eval/gt_loader.py` (reads `start_time`/`end_time`).
- **Alignment is mandatory before trusting any score.** The label timebase MUST match the windowed proxy
  (same fps + frame count). Verify with `ffprobe` (the `largetest_doubles` near-miss is why) ? ideally
  annotate the *exact* proxy the detector ran on (then `video_id` = MD5 matches and alignment is free).
- **`video_id`** = MD5 of the proxy (`backend/utils/hashing.py::compute_video_id`); it keys the WASB
  trajectory cache, the Gemini reference rows, and the golden features.
- **Ingestion** = copy the trajectory + labels to durable paths, write `output/<name>_golden_features.json`
  (`gts` + candidates), append a row to `output/human_lovo_manifest.json`. (`scratch/at_par.py` scores a
  single video vs Gemini + golden; `backend/eval/rally_seg_eval.py` runs the whole corpus.)
- **Path smoke check:** before running corpus evals, run
  `python -m backend.eval.golden_manifest --manifest output/human_lovo_manifest.json --min-count 15`.
  The checker rebases stale absolute paths against sibling `Annotation Setup` folders and `output/`, and
  exits non-zero if any required trajectory or labels file is missing. Add `--write output/human_lovo_manifest.local.json`
  when a resolved local manifest artifact is useful.
- **Guardrails:** owned footage only; **minors excluded**; labels never derived from a paid LLM (Gemini is
  inference/reference only). Commercial-clean.

Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 15)

#	name	sport	rallies	fps	source proxy	video_id	labeled
1	adarsh_avi_singles	badminton (S)	96	59.94	(original 6)	?	pre-06-16
2	kushagra_singles	badminton (S)	32	59.94	(original 6)	?	pre-06-16
3	largetest_doubles	badminton (D)	49	29.97	(original 6)	?	pre-06-16
4	gbaaddy	badminton	48	59.94	(original 6)	?	pre-06-16
5	testlarge_short	badminton	6	29.97	(original 6)	?	pre-06-16
6	mahadevpura_singles	badminton (S)	31	29.97	(original 6)	?	pre-06-16
7	mahadevpura_2	badminton (multicourt)	40	59.94	`?/Roora Request - June 15/[Done] Video 3 ?/mahadevpura-2_proxy.mp4`	`038e0e0915a2`	2026-06-16
8	GX010128	badminton (multicourt)	52	59.94	`?/Golden Label Videos Request - June 15/[Done] Video 13 ?/GX010128.MP4`	`db84f3e65318`	2026-06-16
9	mahadevpura_1	badminton (S, continuous)	10	59.94	`?/[Done] Video 14 - Badminton/mahadevpura-1.MP4`	`38bef75d78cb`	2026-06-17

Labeled, ingestion in progress (WASB trajectory extracting)

name	sport	rallies	fps	source proxy	status
GX030094	badminton	41	**29.97**	`?/Golden Label ? /[Done] Video 6 -

Badminton/GX030094_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
52 | Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `?/[Done] Video 10 - Badminton Multicourt/Badminton BXH 2_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
53
54 ## Source folder & not-yet-labeled
55 - **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\`
56 (+ the earlier `Roora Request - June 15\`). `[Done]` prefix in a dir name = golden-labeled.
57 - **Labeled badminton dirs:** Video 3 (=mahadevpura-2), 6 (=GX030094), 10 (=BXH 2), 13 (=GX010128, a
58 re-label of #8), 14 (=mahadevpura-1). **Video 13 is a duplicate of #8** (already ingested).
59 - **Not yet labeled / other sports:** Video 1, 4, 7, 9, 12 (badminton, pending); Video 2, 8 (table tennis);
60 Video 5, 11 (pickleball). Badminton is the current focus (alpha customers); TT/pickleball labeled later.
61
62 ## How to isolate the golden videos
63 Each row's **source proxy** column is the file to copy into a dedicated golden-videos location. The labels
64 live beside the proxy (`<name>.rallies.csv`) and a durable copy is in `output/<name>.rallies.csv`; the
65 trajectory is in `Annotation Setup/Trajectories/` (older manifests may still say `Collect/Trajectories/`;
66 use the path smoke checker above). Keep this table in sync on every new ingest.
67

12. user (2026-07-06T20:14:13.985Z)

1 # Phase-1 Golden-Set Video Collection Spec
2
3 **Status:** Design v1 · **Date:** 2026-06-02
4 **Companion docs:** [GOLDEN_SET_VENDORING.md](../archives/roora-vendor/GOLDEN_SET_VENDORING.md) (vendor process, ADR-006) · VIDEO_RECORDING_GUIDELINES.md (capture spec the user-facing guide derives from) · scoring in [EVALUATION.md](../EVALUATION.md).
5
6 What footage to gather for the **first round** of golden-set creation so the trained rally segmenter reaches good precision/recall ? and so the eval number actually predicts production. Golden-set creation is **iterative**: expect multiple rounds, expanding toward whatever axis round-1 metrics show is weakest.
7
8 ---
9
10 ## 1. The principle that dominates everything: match the deployment distribution
11
12 The CV literature is consistent: **models trained on broadcast footage fail to generalize to self-recorded amateur video** ? broadcast uses a "straight, fixed camera with minimum variability," while real non-elite footage has limited coverage, occlusions, and inconsistent viewpoints ([team-sports survey](https://link.springer.com/article/10.1007/s10462-024-10934-9), [unstructured-video event detection](https://arxiv.org/pdf/2002.08097)). This is the **same failure mode as ADR-004** (broadcast cameras *track players*, so player pixel-motion inverts during rallies).
13
14 Our product ingests **users' own match recordings** (a fixed phone/GoPro on a tripod). Therefore:
15
16 - The golden set ? **especially the held-out eval split** ? must be dominated by **static, self-recorded, full-court** footage that mirrors real users.
17 - **ShuttleSet / broadcast** footage is fine as *training* augmentation and as a contrast set, but must be **minimal in eval**. Measuring precision/recall on broadcast yields a number that does **not** predict production.
18
19 ---

2. Primary recording profile (also the basis of the user-facing capture guide)

Grounded in the badminton-CV setup literature (camera ~2 m behind the baseline, ~4.5 m elevated, ~30° down-tilt, whole court + headroom above the net ? [source](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)):

Setting	Spec	Why
Mount	Fixed tripod; **no pan/zoom** during play	Avoids the ADR-004 camera-tracking trap
Position	Behind one baseline/end, centered	Full-court, symmetric view
Height	**3?5 m** elevated (~4.5 m ideal)	De-occludes the far player & net
Tilt	**~30° down**	Optimal coverage, low keystoning
Framing	Entire court **+ headroom above the net**; players never leave frame	Detection + court-relevance need it
Lighting	Even; no backlight/glare	Reduces blur & false motion

Most of round 1 should use this profile; a minority deliberately varies it (§4) to probe robustness.

---

3. Resolution & frame rate (per sport)

A controlled study: table-tennis ball detection was **22.5% @ 30 fps ? 93.2% @ 60 fps** (120 fps ? 60 fps) ([Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6)); OpenTTGames records at 120 fps. The ball/shuttle is the camera-invariant rally signal (our WASB substrate, ADR-001), so fps is **not** a nicety for fast sports.

Sport	Min resolution	Recommended fps	Note
Badminton	1080p (720p floor)	30 ok, **50/60 better**	Shuttle decelerates fast; 60 sharpens smashes
Pickleball	1080p	**30?60**	Ball slower than TT; 30 likely fine, 60 safer
Table tennis	1080p	**60 min (120 ideal)**	30 fps collapses ball detection

> These are **literature-backed starting points**; the published minimums get finalized empirically against our golden set (VIDEO_RECORDING_GUIDELINES.md §3).

---

4. The Phase-1 diversity matrix

Cover the **high-impact axes** now; don't chase rare conditions in round 1. "Guards" = whether the axis mostly protects **P**recision (not calling dead-time a rally) or **R**ecall (not missing rallies).

Dimension	Vary across?	Guards	Round-1 target
Sport	badminton, pickleball, table tennis	both	all 3
Format	singles **and** doubles (badminton, pickleball)	R	both for those 2
Camera profile	mostly primary-static; 1?2 side-elevated; 1?2 lower/phone-height; **1?2 broadcast (contrast only)**	both	~80% static / ~20% variants
Venue / court color	wood, green, blue synthetic	P	?3 venues
Lighting	bright gym, dim club, mixed/backlit	both	?3 conditions
Background clutter	empty hall vs people moving behind court	**P** (crowd motion ? false positives)	both
Players	skill, attire colors, body types, L/R-handed, skin tone/age	R	spread; don't reuse one pair
Rally edge cases	service faults, lets/replays, 1?2-shot & long rallies, net-cords, coaching pauses, warm-up	boundary **P/R**	ensure present
Resolution/fps ablation	same match re-encoded 480/720/1080p; 30 vs 60 fps	calibration	1 matched set/sport

---

5. Split it so the precision/recall number is honest (no train/eval leakage)

The #1 video-ML mistake: the **same match in train and eval inflates the score** ([data-leakage guide](https://towardsdatascience.com/data-science-mistakes-to-avoid-data-leakage-e447f88aae1c/)).

- **Split by match / recording session ? never by clip.** All clips from one match go entirely to train *or* eval.
- **Hold out whole *conditions*** (a venue, a camera profile, a player pair the model never saw) so eval measures generalization, not memorization.
- Reserve **~? of each sport as a deployment-like held-out eval set** (static, self-recorded). Broadcast/ShuttleSet may enter train but should be **minimal in eval**.

> **Guardrails ? TODO (design with the user):** train and eval live in **separate catalogs** today, kept apart by convention. We want hard guardrails ? permissions / explicit approval / re-confirmation at ingest & export ? so a video can **never** silently land in both. Tracked in BACKLOG.md; to be designed in a later session.

---

6. Concrete "round 1" shopping list (per sport)

Aligns with the 12?15-videos/sport scale plan; the 3-clip Roora calibration is a subset of this.

- **8 static-primary** clips spanning venues / lighting / skill (?4 singles, 4 doubles where applicable)
- **2 camera-variant** clips (side-elevated, lower height) ? robustness probe
- **1 resolution/fps ablation set** (one match @ 480/720/1080p, 30 & 60 fps)
- **1 broadcast contrast** clip (ShuttleSet) ? *train/contrast only*
- Each ~2?4 min, **deliberately including the edge cases** above ? **~12 clips/sport**; hold out **4 whole matches** for eval.

---

7. Deliberately defer to later rounds

Outdoor courts, phone-handheld/jitter, extreme angles, 4K, unusual net/court variants, crowd-heavy tournaments. Add these once round-1 precision/recall shows the weakest axis, then expand toward it (find the quality "knee" ? VIDEO_RECORDING_GUIDELINES.md §3).

---

8. Two cautions specific to precision (the stated goal)

- **Backgrounds with people moving** behind the court are the biggest false-positive driver. Include some so the model learns to reject them ? and make sure the annotation guideline's "ignore all dead time" rule is airtight.
- **Warm-up / knock-up / practice swings** look like rallies. Capture some, and confirm the vendor **excludes** them (the Roora guide now has an explicit rule + a how-to-spot-it tip). Early precision is bounded more by **guideline / inter-annotator consistency (IAA)** than by raw data volume.

---

9. Sources

- [Team-sports action-recognition survey ? Springer 2024](https://link.springer.com/article/10.1007/s10462-024-10934-9)
- [Unstructured-video event detection ? arXiv 2002.08097](https://arxiv.org/pdf/2002.08097)
- [Badminton service-height CV setup (camera 4.5 m / 30°) ? PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)

~~~~
- [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
- [Table-tennis frame-rate vs detection accuracy ? Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6)
- [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)
- [Action-spotting / segmentation survey ? arXiv 2505.03991](https://arxiv.org/html/2505.03991v2)
- [Data-leakage in train/test splits](https://towardsdatascience.com/data-science-mistakes-to-avoid-data-leakage-e447f88aae1c/)

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13. user (2026-07-06T20:14:14.794Z)

~~~~
# Golden-Set Subproject ? Design & Implementation Overview

**Status:** In progress · **Date:** 2026-05-30
**Companion docs:** design rationale + vendor shortlist in [GOLDEN_SET_VENDORING.md](../archives/roora-vendor/GOLDEN_SET_VENDORING.md) (ADR-006); scoring mechanics in [EVALUATION.md](../EVALUATION.md).

This is the **implementation tracker** for turning the golden-set/vendor *design* into shipped code. It's split into small, independently-reviewable PRs. The whole subproject is **vendor-agnostic by construction** ? only one slice (GS-3, `HumanVendorProvider`) depends on which vendor you pick offline; everything else can land first.

> **Update (2026-06-02):** Vendor selected for the **initial pilot ? Roora** (Bengaluru). The **annotation guideline, intake-manifest template, and NDA** are authored in the collector repo `docs/annotation/`, and the collector's `import-local` ingestion now covers the **3 pilot sports** (badminton, pickleball, table tennis). On the indexer side, GS-1 (`file_provider.py`), GS-2 (`backend/eval/regression.py` + `run_regression.py`), and the GS-4 oracle scaffold (`automated_provider.py`) have landed; the remaining slice is the vendor-dependent **GS-3 `HumanVendorProvider`**, parked until the Roora pilot confirms quality.

---

## Architecture in one picture

```
            video_ref ??? AnnotationProvider.submit()/poll()/fetch() ??? List[(start,end)]   ground truth
                                  ?          ?           ?                              ?
                           FileProvider  HumanVendor   Automated(oracle)                  ?
                             (GS-1)        (GS-3)         (GS-4)          backend.eval.metrics.score()  ??? existing, unchanged
                                                                                          ?
     production pipeline ??? DB ??? backend.eval.harness.get_predictions() ??? List[(start,end)]   predictions
                                                                                          ?
                                                              regress(video, tier) (GS-2) compares the two
```

**The keystone:** a provider's `fetch()` returns the **same `List[(start, end)]`** that `eval.annotations.load_annotations()` already returns and `eval.metrics.score()` already consumes. So one scorer grades all three of: product-vs-GT, vendor-vs-answer-key, and annotator-vs-annotator (IAA). No new scoring code ? only new *sources* of intervals.

---

## PR breakdown

| Slice | Branch | Scope | Depends on | Vendor-dependent? |
|---|---|---|---|---|
| **GS-1** | `feat/annotation-provider-registry` | `AnnotationProvider` ABC + registry + `FileProvider` + tests | ? | No |
| **GS-2** | `feat/regression-runner` | `regress(video, tier, tolerance)` over existing
~~~~

```
harness + `baseline.json` snapshot/compare + CLI | GS-1 | No |
| **GS-3** | `feat/human-vendor-provider` | `HumanVendorProvider` (submit/poll/fetch against the chosen vendor's API or file-drop), DPA/secure-link handling | GS-1, **your vendor pick** | **Yes** |
| **GS-4** | `feat/automated-oracle-provider` | `AutomatedProvider` fast-tier oracle ? **interface stub** (injected labeler + fake; concrete model = backlog) | GS-1, GS-2 | No |
| **GS-5** | `feat/regression-triggers` | Wire triggers: CI gate, on-demand CLI, scheduled run | GS-2, GS-4 | No |

Each row is one PR. GS-1 ? GS-2 ? GS-4 ? GS-5 can proceed **now**, independent of vendor selection. GS-3 is parked until you've chosen a vendor (then it's a thin adapter behind the GS-1 interface).

---

## Slice details

### GS-1 ? Provider registry + FileProvider  ? (this PR)
- `backend/annotations/base.py` ? `AnnotationProvider` ABC (`submit`/`poll`/`fetch` + `annotate()` convenience) and `annotation_registry`, mirroring `backend/providers` and `backend/indexer` registry patterns.
- `backend/annotations/file_provider.py` ? `FileProvider`: the "job" is a CSV that already exists (formalizes today's manual/ShuttleSet flow); reuses `eval.annotations.load_annotations`.
- Tests assert the registry contract **and** the keystone (provider output is directly scorable by `metrics.score()`).
- **Design choice:** `fetch()`/`annotate()` raise loudly on a missing/incomplete job rather than returning `[]`, so a broken label source can't silently masquerade as "0 rallies" and skew a regression to green.

### GS-2 ? Regression runner  (next)
`regress(video, tier="fast", tolerance=0.05)`: get GT from `annotation_registry.get(tier)`, read predictions via the existing `eval.harness`, score with the existing `metrics`, compare against a per-video `baseline.json`, and flag a regression when precision/recall drop beyond tolerance. Pure orchestration over existing parts + a small baseline store. CLI: `python run_regression.py --video ... --tier file`.

### GS-3 ? HumanVendorProvider  (after your vendor pick)
A concrete provider that submits a (licensed/owned-only) video to the chosen vendor and fetches the returned rally CSV. Carries the privacy controls from the design doc (secure time-limited links, delete-on-delivery, DPA). Thin adapter ? the interface already exists from GS-1.

### GS-4 ? AutomatedProvider (fast-tier oracle)  ?? oracle rule
An independent strong labeler for the fast/CI tier. **Must be a different model lineage than whatever production uses** (else shared blind spots ? false green). A fast-tier failure is a *smoke alarm* that escalates to the GS-3 human tier, never an authoritative verdict on its own.

### GS-5 ? Regression triggers
Wire `regress()` into CI (pre-merge gate on pipeline changes), an on-demand CLI (the "surprise regression"), and an optional scheduled run. Includes a deliberate re-snapshot path so legitimate production improvements can rebaseline instead of raising false regressions.

---

## Conventions
- **Branch per slice**, one PR each, tests green before PR, full suite must stay green.
- New runtime deps: **none** through GS-2 (GS-3 may add a vendor SDK; GS-4 may reuse the existing `google-genai` for a Gemini oracle).
- All ground-truth video stays **licensed/owned** ? see the privacy section of the design doc.
```

14. assistant (2026-07-06T20:14:19.150Z)

I now have all six files read fully. Let me note one important discrepancy to verify: the label CSV format differs between docs. `EVALUATION.md` says `start,end`; `GOLDEN_VIDEOS.md` says `rally_number,start_time,end_time,ending_reason,sport,shots_count`; `DATA_IN_GCS.md` uses `<clip>.rallies.csv`. Let me check the actual `gt_loader.py` to resolve which columns are authoritative, since the question asks for exact CSV format.

15. user (2026-07-06T20:14:19.788Z)

backend\eval\gt_loader.py

16. user (2026-07-06T20:14:19.993Z)

backend\eval\annotations.py

17. assistant (2026-07-06T20:14:22.492Z)

Both exist. Let me read them to resolve the CSV column discrepancy authoritatively.

18. user (2026-07-06T20:14:23.412Z)

```
"""Canonical ground-truth interval loader ? THE one GT/golden-CSV reader (ADR-008 prereq).

The repo historically carried two incompatible loaders:
- ``annotations.load_annotations``: positional ``start,end`` (the collector's
  ``curate export`` format), raises on bad rows;
- ``calibrate_wasb.load_golden_gts``: DictReader requiring ``start_time/end_time``
  (the rally-annotator golden format), silently skipping bad rows ? fed 7 modules.

A file in one convention fed to the other loader yielded zero/garbage intervals with no
error. This module accepts BOTH conventions (header-mapped, else positional), so every
quantity the ship-gate depends on (gt_hash, LOVO scores, the future consent fingerprint)
flows through one reader. Both legacy entry points now delegate here.

Accepted shapes (one rally per row; blank lines and ``#`` comments ignored):
- header with ``start``/``end`` OR ``start_time``/``end_time`` columns (any other
  columns ? rally_number, ending_reason, sport, ? ? are ignored);
- no header: columns 0,1 positionally.
Timestamps: decimal seconds or ``MM:SS`` / ``HH:MM:SS`` (``annotations.parse_timestamp``).
"""

import csv
import logging
from typing import List, Optional, Tuple

from backend.eval.annotations import parse_timestamp

logger = logging.getLogger(__name__)

Interval = Tuple[float, float]

_START_NAMES = ("start", "start_time")
_END_NAMES = ("end", "end_time")


def _header_columns(row: List[str]) -> Optional[Tuple[int, int]]:
    """If `row` is a header naming start/end columns, return their indices, else None."""
    cells = [(c or "").strip().lower() for c in row]
    start_idx = next((i for i, c in enumerate(cells) if c in _START_NAMES), None)
    end_idx = next((i for i, c in enumerate(cells) if c in _END_NAMES), None)
    if start_idx is not None and end_idx is not None:
        return start_idx, end_idx
```

```
    return None


def load_gt_intervals(csv_path: str, *, strict: bool = True) -> List[Interval]:
    """Load GT rally intervals from either golden-CSV convention.

    strict=True (corpus/gate use): raise ValueError with file:line on any malformed row
    (bad timestamp, end <= start, negative start) so labeling mistakes fail loudly.
    strict=False (legacy tuning-driver behavior): skip bad rows, but LOG how many were
    skipped ? a silently shrinking GT set flatters precision.

    Returns intervals sorted by start; overlapping GT intervals are logged as a warning
    (they break one-to-one matching and usually indicate a labeling error).
    """
    intervals: List[Interval] = []
    skipped = 0
    cols: Optional[Tuple[int, int]] = None  # (start_idx, end_idx); None until detected
    saw_header = False

    with open(csv_path, newline="", encoding="utf-8-sig") as f:
        for lineno, row in enumerate(csv.reader(f), start=1):
            if not row or not (row[0] or "").strip() or row[0].strip().startswith("#"):
                continue
            if cols is None and not saw_header:
                header = _header_columns(row)
                if header is not None:
                    cols = header
                    saw_header = True
                    continue  # consume the header row
                cols = (0, 1)  # headerless: positional start,end
            assert cols is not None  # every branch above assigns it
            si, ei = cols
            try:
                if len(row) <= max(si, ei):
                    raise ValueError(
                        f"expected at least {max(si, ei) + 1} columns, got {row!r}"
                    )
                start = parse_timestamp(row[si])
                end = parse_timestamp(row[ei])
                if start < 0:
                    raise ValueError(f"negative start ({start})")
                if start >= end:
                    raise ValueError(f"start ({start}) >= end ({end})")
            except ValueError as e:
                if strict:
                    raise ValueError(f"{csv_path}:{lineno}: {e}") from e
                skipped += 1
                continue
            intervals.append((start, end))

    if skipped:
        logger.warning(
            "%s: skipped %d malformed GT row(s) (strict=False)", csv_path, skipped
        )

    intervals.sort(key=lambda iv: iv[0])
    for (s1, e1), (s2, e2) in zip(intervals, intervals[1:]):
        if s2 < e1:
            logger.warning(
                "%s: overlapping GT intervals (%.3f-%.3f vs %.3f-%.3f) ? "
                "labeling error? Overlaps break one-to-one matching.",
                csv_path,
                s1,
                e1,
                s2,
```

```
                    e2,
                )
        return intervals
```

19. assistant (2026-07-06T20:15:48.191Z)

I now have everything needed to resolve the CSV format discrepancy authoritatively. I have all the facts. Here is the consolidated report.

Report: Golden eval videos, labels, quick-run clips, and F1/tolF1 definitions

Five docs read in full plus the authoritative loader `backend/eval/gt_loader.py`. Note upfront: this repo describes **two parallel worlds** ? (1) a legacy **local-filesystem / SQLite-DB / OneDrive-Drive** world (the older docs), and (2) a newer **GCS bucket** world (`gs://khelsutra-rally-corpus`) that is mid-migration. I flag which is which throughout.

(a) Golden-video locations, filenames, and GCS URIs

GCS bucket (the current source-of-truth target)

Evidence: `docs\DATA_IN_GCS.md`
- **Bucket:** `gs://khelsutra-rally-corpus` (single corpus bucket), project `gen-lang-client-0306026826`, region `asia-south1`. ? lines 30-33.
- **Canonical layout contract** ? lines 48-55:
  - `videos/` holds `<clip>.mp4` ? the video prefix.
  - `labels/` holds `<clip>.rallies.csv` ? golden rally labels.
  - `trajectories/` holds cached WASB trajectory CSVs (the GPU-free eval input, consumed via `manifests/golden.json`).
  - `manifests/golden.json` ? the eval manifest (15 clips, `gs://` trajectory + label paths) ? lines 207-209.
- **Example URIs** (from the worker command block, lines 109-117): `gcs://khelsutra-rally-corpus/videos/<clip>.mp4`, out-URI `gcs://khelsutra-rally-corpus`.
- **? Reality vs. contract:** only the **metadata** (15 trajectories + 15 canonical `labels/<date>_<name>.rallies.csv` + `manifests/golden.json`) actually landed in the bucket (Phase-1, 2026-06-22). The **videos are almost entirely NOT migrated** ? "**1/15** demo'd (`mahadevpura_1`); 14 remain" ? lines 275. As of the pre-Phase-1 snapshot, `videos/` held only 3 objects: `GX030094_proxy.mp4`, `mahadevpura-1.MP4`, `mahadevpura_smoke_180s.mp4` ? line 77.
- **Owned-source video MD5 registry** (durable manifest, not the videos themselves): `sports-data-collector/data/golden_source_videos.json` (six owned golden MP4s) ? lines 139, 218-220.

Local / Drive world (where the actual golden video proxies live today)

Evidence: `docs\data_pipeline\GOLDEN_VIDEOS.md`
- **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos Request - June 15\` (+ earlier `Roora Request - June 15\`). `[Done]` prefix = golden-labeled. ? line 55.
- **Actual per-clip source-proxy paths** (the "source proxy" column, lines 43-45, 51-52):
  - `mahadevpura_2` ? `?/Roora Request - June 15/[Done] Video 3 ?/mahadevpura-2_proxy.mp4`, video_id `038e0e0915a2`
  - `GX010128` ? `?/Golden Label Videos Request - June 15/[Done] Video 13 ?/GX010128.MP4`, video_id `db84f3e65318`
  - `mahadevpura_1` ? `?/[Done] Video 14 - Badminton/mahadevpura-1.MP4`, video_id `38bef75d78cb`
  - `GX030094` ? `?/[Done] Video 6 - Badminton/GX030094_proxy.mp4`
  - `Badminton BXH 2` ? `?/[Done] Video 10 - Badminton Multicourt/Badminton BXH 2_proxy.mp4`
- **Phase-2 migration source table** (`docs\DATA_IN_GCS.md` lines 168-170): 7 Drive proxies (`mahadevpura_1/2`, `GX010128`, `GX020094`, `GX030094`, `Badminton_BXH_2`, `Boxhill_Doubles`, ~13.5 GB) + 8 F:-archive full MP4s (`GX010137`, `GX010141`, + original-6: `adarsh_avi_singles`, `kushagra_singles`, `largetest_doubles`, `gbaaddy`, `testlarge_short`, `mahadevpura_singles`, ~50 GB). F: root: `F:\[Khelsutra] GoPro Hero Black 12\GoPro Recorded Videos`.

De-attributed regression fixtures (a third, video-free artifact ? not videos)

Evidence: `docs\GOLDEN_REGRESSION_FIXTURES.md`
- Tier-0 fixtures live in-repo at `eval_baselines/fixtures/` (tracked) ? line 87. Six real fixtures `fx_r01`?`fx_r06` (~1.9 MB total), de-attributed per-frame `trajectory.csv`, both 29.97 & 59.94 fps ? lines 94, 159, 174. These are **not** playable videos; they are frozen numeric trajectory streams for cross-machine regression. The surrogate?source name map is kept **off-git** (owner-held).

(b) Ground-truth labels: location, CSV format, and video?label pairing

Location
- **GCS:** `labels/<YYYY-MM-DD>_<name>.rallies.csv` ? `docs\DATA_IN_GCS.md` line 199 (naming LOCKED decision; `<YYYY-MM-DD>` = label authoring date = earliest mtime).
- **Local durable copies:** `output/<name>.rallies.csv`, plus beside the proxy as `<name>.rallies.csv` ? `GOLDEN_VIDEOS.md` lines 64-66.
- **Manifest tying it together:** `output/human_lovo_manifest.json` (local, n=15) and `gs://?/manifests/golden.json` (GCS, 15 clips).

CSV format ? ? TWO conventions, both accepted (this is a real, load-bearing discrepancy)
The docs disagree, and I resolved it against the authoritative loader `backend\eval\gt_loader.py`:
- **`EVALUATION.md` (lines 27-31):** positional/headered `start,end` ? decimal seconds or `MM:SS`/`HH:MM:SS`, header optional/auto-detected. This is the collector's `curate export` format.
- **`GOLDEN_VIDEOS.md` (lines 15-16):** rally-annotator CSV with columns `rally_number,start_time,end_time,ending_reason,sport,shots_count` (decimal seconds), ingested by `backend/eval/gt_loader.py` which reads `start_time`/`end_time`.
- **Authoritative resolution ? `gt_loader.py` accepts BOTH** (it was written specifically to unify two historically-incompatible loaders ? docstring lines 3-18). Rules (lines 31-32, 45-90): it matches a header naming `start`/`start_time` (start col) and `end`/`end_time` (end col); **all other columns ? `rally_number`, `ending_reason`, `sport`, `shots_count` ? are ignored**; if no header, uses columns 0,1 positionally. `strict=True` (corpus/gate) raises `file:line` on any bad row; `strict=False` (legacy) skips+logs. So the two "different" formats are the same start/end pair with extra ignored columns.

Video ? label pairing convention
- **Filename convention:** `<name>.rallies.csv` pairs with the clip named `<name>` ? confirmed in `DATA_IN_GCS.md` layout (`videos/<clip>.mp4` ? `labels/<clip>.rallies.csv`, lines 49-50) and `GOLDEN_VIDEOS.md` line 64. GCS labels additionally carry a date prefix; `backend/worker/__main__.py::_label_clip_name()` strips both the `.rallies.csv`/`.csv` suffix **and** a leading `^\d{4}-\d{2}-\d{2}_` prefix so `2026-06-22_GX020094.rallies.csv` resolves to bare `videos/GX020094` ? `DATA_IN_GCS.md` lines 238-246.
- **Strong keying (MD5):** `video_id` = MD5 of the proxy via `backend/utils/hashing.py::compute_video_id`; it keys the WASB trajectory cache, Gemini reference rows, and golden features ? `GOLDEN_VIDEOS.md` lines 17-19. The eval CLI keys the DB by file MD5 (`--video <file>` computes it, or `--video-id <md5>`) ? `EVALUATION.md` lines 51-57.
- **Alignment caveat:** label timebase must match the windowed proxy (same fps + frame count); verify with `ffprobe` ? the `largetest_doubles` "near-miss" is cited as why ? `GOLDEN_VIDEOS.md` lines 18-19.

(c) 2-3 specific clips for a quick numbers run (with resolution / fps)

Best candidates, drawn from the ingested corpus table in `GOLDEN_VIDEOS.md` (lines 37-52) and the R2 preview clips:

Clip	Sport / type	Rallies	fps	Notes / evidence
GX010128	badminton, multicourt	52	**59.94**	Has a fully worked tolF1 preview (see (d)); video_id `db84f3e65318`. `GOLDEN_VIDEOS.md:44`; `R2_EVAL_METRIC_DESIGN.md:108-109`.
mahadevpura_2	badminton, multicourt	40	**59.94**	The primary n=2 preview clip;

video_id `038e0e0915a2`. `GOLDEN_VIDEOS.md:43`; `R2_EVAL_METRIC_DESIGN.md:106-107`. |
| **mahadevpura_1** | badminton (S, continuously-active blob) | 10 | **59.94** | Small (10 rallies), fast; reported as its own stratum. It's also the **only clip already in the GCS `videos/` bucket**. video_id `38bef75d78cb`. `GOLDEN_VIDEOS.md:45`; `DATA_IN_GCS.md:275`. |
| (alt, smallest) **testlarge_short** | badminton | 6 | 29.97 | Fewest rallies of all; but "original 6" with no video_id/GCS proxy recorded. `GOLDEN_VIDEOS.md:41`. |

Resolution: the docs give an explicit numeric fps per clip but **do not state per-clip pixel resolution** in these registries. The only resolution guidance is the spec target ? Badminton **1080p** (720p floor), 30 ok / 50-60 better ? `GOLDEN_SET_PHASE1_VIDEOS.md` lines 44-46; the Drive proxies are noted as "already 1080p" (`DATA_IN_GCS.md` line 180). **Per-clip actual resolution is an open item** (see flags below). The named 3-clip "Roora calibration" subset is mentioned but not enumerated by name ? `GOLDEN_SET_PHASE1_VIDEOS.md` line 84.

`n2` ground-truth clips used across the metric design = **mahadevpura-2 and GX010128** (`R2_EVAL_METRIC_DESIGN.md` line 34, table lines 106-109) ? the cleanest pair for a quick numbers run.

(d) F1 / tolF1 definition and the ? tolerance values

Two distinct metrics coexist. Do not conflate them.

1. tIoU-based F1 (the current gate) ? `docs\EVALUATION.md`
- **Matching:** predictions and GT are `[start, end)` intervals; a prediction matches a GT rally when their **temporal IoU (overlap ÷ union) ? threshold**. Matching is **greedy, one-to-one** (highest-IoU pairs claimed first) ? lines 112-118, 5-line "How matching works" section.
- **IoU threshold ?:** default **0.5** (`--iou`, default `0.5`); optional PR sweep 0.1-0.9 ? lines 56, 65-67.
- **P / R / F1:** precision = TP/(TP+FP), recall = TP/(TP+FN), **F1 = their harmonic mean** ? lines 87-89. Also reports mIoU (mean temporal IoU over matched pairs) and dStart/dEnd (mean absolute boundary error in seconds).
- **Level:** rally-level (per-video `[start,end)` rally intervals). Scores predictions already stored in the SQLite DB; deterministic, no API calls ? lines 3-7.
- **Single-video command** (`EVALUATION.md` lines 51-57):
  - `python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv`
  - or `python run_eval.py --video-id <md5> --annotations rallies.csv --stage both --iou 0.5`
  - **Inputs needed:** the video must already be processed so detections are in the DB (harness does **not** run the pipeline), plus the ground-truth `rallies.csv`. Keyed by file MD5.

2. tolF1 / F1@? (the newer "R2" tolerance metric) ? `docs\R2_EVAL_METRIC_DESIGN.md`
- **Definition (the exact difference from tIoU-F1):** instead of an IoU-overlap gate, a predicted window matches a golden rally **iff `|?start| ? ?_start` AND `|?end| ? ?_end`** (asymmetric boundary tolerance) ? line 52. Assignment is **strict 1:1 Hungarian** (`scipy.optimize.linear_sum_assignment`), *not* greedy/overlap ? lines 62-64. Then `P@? = matched / n_pred`, `R@? = matched / n_ref`, `F1@?` = harmonic mean ? line 69. Reported as a **decomposition** (P@?, R@?, F1@?, split start/end boundary-error distributions, over-seg guard, ?-sweep), never one number ? lines 68-88.
- **? values ? in SECONDS** (temporal, not pixels):
  - Provisional locked pair: **`?_start = 2.0 s`, `?_end = 1.5 s`** (asymmetric ? forgive the fuzzy start, keep the rally-end honest) ? lines 53-56, 95-96, 183-186.
  - ?-sweep set (symmetric, for the trend line): **`? ? {1.5, 2.0, 2.5, 3.0}`** seconds ? line 84.
  - These ? are explicitly **provisional pending an IAA (inter-annotator agreement) study**; final `?_start`/`?_end` should be set to ~the 90th-percentile pairwise human `|?start|`/`|?end|` ? §2.4, lines 89-96.
- **tolF1 vs F1 relationship:** `tolF1@~3.0 ? tIoU-F1@0.5` ? the tolerance sweep "contains tIoU as a point" (continuity with the frozen baseline) ? lines 105, 112-113.
- **Status / which is the gate:** R2 is **IMPLEMENTED but default-additive** ? `backend/eval/tolerance_metrics.py` landed and `rally_seg_eval` reports the R2 block **alongside** tIoU; **tIoU@0.5 remains the gate** pending an ablation experiment ? lines 2-6, 147-149, 187-195. So today the authoritative headline number is still tIoU-F1@0.5; tolF1 is reported next to it.

- CLI flags: `--tau-start` / `--tau-end` landed on `rally_seg_eval`; the ?-sweep helper `tolerance_metrics.tolerance_sweep` exists but is not yet surfaced as a CLI flag ? lines 148, 167.

Open questions / things NOT found (flagged)

1. **Per-clip pixel resolution is not recorded** in any of these docs. The registries give fps per clip but only spec-level target resolution (1080p badminton). `DATA_IN_GCS.md` line 191 even warns that fps/frame_width must be **re-probed from the bucket videos** in Phase-3 ("don't trust the copied values"). If exact resolution per clip is needed, it must be `ffprobe`'d from the actual proxy ? it's not documented.
2. **The named 3-clip "Roora calibration" subset** is referenced (`GOLDEN_SET_PHASE1_VIDEOS.md:84`) but the three specific clip names are never enumerated. My recommended quick-run trio (GX010128, mahadevpura_2, mahadevpura_1) is inferred from the ingested-corpus table + the R2 preview, not from an explicit "quick set" list.
3. **CSV-format inconsistency across docs is real but harmless:** `EVALUATION.md` says `start,end`; `GOLDEN_VIDEOS.md` says the 6-column `rally_number,start_time,end_time,ending_reason,sport,shots_count`. `backend/eval/gt_loader.py` reconciles both (reads start/end under either header, ignores extra columns). Cite `gt_loader.py:1-18,31-90` as authoritative if this matters downstream.
4. **Video availability gap:** despite the GCS "source-of-truth" framing, only 1 of 15 golden videos is actually in `gs://?/videos/` today. For a live numbers run you currently need the local Drive proxies (`G:\My Drive\...`) or the F: archive, plus a processed DB ? the GCS eval path (Phase-3) can't fetch `gs://` yet (`DATA_IN_GCS.md:186-194`, blocker §7b#2).
5. **No `.MP4` ? `.rallies.csv` literal example** is given verbatim, but the convention is unambiguous from the layout table: `videos/<name>.mp4` pairs with `labels/<name>.rallies.csv` (GCS adds a `<date>_` prefix on the label, stripped at join time).

All paths above are quoted from the docs; file:line regions are cited inline for each fact.